Financial Frictions, FX Reserves, and Exchange Rate Management with Local-Currency Debt

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Abstract

Emerging economy central banks often hold large FX reserves while residents carry substantial local-currency-linked external liabilities. With financial frictions creating interest parity gaps, such opposing positions imply a carry cost. In a small open economy with intermediation frictions and inherited local-currency debt, we study optimal reserve and exchange rate policies, explaining why debtor economies may retain reserves rather than netting out costly gross positions. While policy can eliminate costly intermediation by deploying reserves and appreciating the real exchange rate, doing so creates a general equilibrium revaluation effect which raises the real burden of local-currency obligations. Optimal policy retains reserves, accepting some intermediation to avoid a larger revaluation loss. This policy is time-inconsistent: a discretionary central bank prefers stronger ex-post appreciation; a time-consistent equilibrium features more reserve retention and larger interest parity gaps. Revaluation costs restrain reserve deployment; when large, optimal policy retains more reserves during disruptions than in normal times.

Keywords: Foreign exchange reserves, financial intermediation frictions, real exchange rate, debt revaluation, local currency debt, interest parity gaps, time-inconsistency.

JEL Classifications: E21, F31, F32, F34, G15.

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1 Introduction

Many emerging economy central banks hold large foreign exchange (FX) reserves while domestic residents carry sizable external liabilities. Figure 1 illustrates this pattern: in 2022 the median country in a sample of large net debtor emerging economies held reserves of about 16 percent of GDP while domestic residents contemporaneously maintained net external liabilities of about 45 percent of GDP. Central bank reserves, typically denominated in the US dollar or other major currencies, are low-yield assets. In contrast, a large share of resident liabilities pass through global financial intermediaries, creating significant financing costs over and above the world interest rate.

These global intermediaries borrow in US dollars from international asset markets and lend to emerging market borrowers in domestic currency. Consequently, a substantial and growing portion of external liabilities in emerging economies are now denominated in domestic currency or are effectively linked to domestic currency through derivatives. However, financial intermediaries face significant balance sheet constraints that limit their ability to intermediate between domestic borrowers and international lenders. Recent empirical work documents persistent interest parity deviations and attributes these to financial intermediation frictions. These deviations imply that accessing international asset markets is costly: there is a wedge between the interest rate paid by domestic borrowers and the world interest rate.

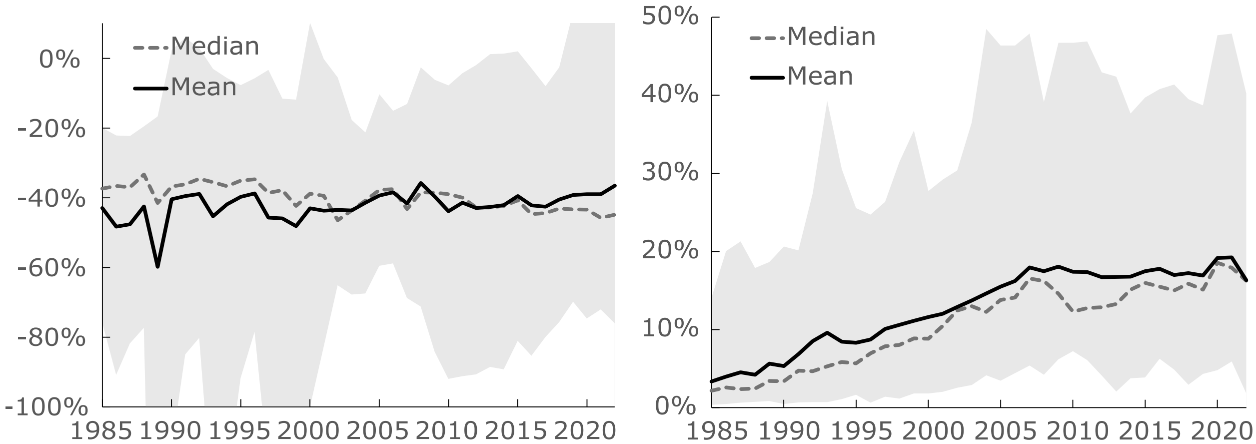
Given this wedge, holding low-yield reserves while residents hold costly external liabilities imposes a carry cost on the economy. Moreover, this carry cost is compounded if the interest parity wedge is endogenous and increasing in the size of the private sector's external liabilities. If the central bank were to reduce its reserves, the private sector's demand for external liabilities would fall. Then, by maintaining these gross positions, the central bank forces the private sector to stay further up the supply curve of international capital. This raises the cost of credit not just for the marginal unit of debt, but for all inframarginal units as well. This raises a natural question: why do debtor economies not use reserves to pay down their external liabilities and thereby eliminate this carry cost?

The answer lies in the valuation effects inherent in a world of domestic-currency-linked liabilities. In the presence of inherited domestic currency claims held by foreign creditors, using reserves to minimize costly external borrowing is not a mechanical debt-reduction exercise. Because reserve operations affect the real exchange rate, they trigger a general equilibrium revaluation effect on the stock of existing liabilities. Specifically, while liquidating reserves induces a real exchange rate appreciation that decreases private external borrowing, this same appreciation also revalues the stock of pre-existing domestic currency liabilities, increasing their real burden in tradeable units. The central bank thus faces a fundamental trade-off: it must weigh the marginal benefit of reducing the carry cost against the cost of allowing a higher real debt burden. Given this, the optimal policy may involve retaining reserves, accepting a persistent carry cost to avoid a costly revaluation of existing obligations.

In a frictionless world where interest parity holds, Ricardian equivalence renders reserve operations irrelevant for real exchange rate management. However, in the presence of financial frictions this equivalence breaks and reserve operations can influence the economy's intertemporal consumption profile by moving the real exchange rate. Therefore, a reserve management policy that accounts for a costly intermediation wedge and revaluation effects has macroeconomic implications beyond optimal asset positions, i.e., it influences the real tradeable resources that an economy can access and consume. We develop a general equilibrium framework that captures these interactions, allowing us to characterize the optimal reserve and exchange rate management policies for debtor economies in the presence of domestic-currency-linked external claims whose real value can be influenced by the central bank's actions. Our framework shows that the presence of revaluation costs imposes a fundamental constraint on how a country can deploy its reserves, not only to mitigate the carry cost of its debt but also to manage the economy's intertemporal consumption path.

We present a model of a pure exchange small open economy with households, international financial intermediaries, and a central bank. Capital mobility is imperfect due to financial intermediation frictions in the spirit of Gabaix and Maggiori (2015). Financial markets are segmented: households access only domestic markets trading bonds denominated in domestic currency, while intermediaries access both domestic and international markets. Crucially, these intermediaries face balance sheet constraints that generate an endogenous failure of interest parity. They demand a premium to absorb domestic-currency bonds, creating a wedge between the domestic and world real interest rates; the amount of assets supplied by them is increasing in the size of this wedge. Since intermediaries are externally owned, this wedge represents a resource cost for the domestic economy in the form of payments to foreign creditors over and above the world interest rate.

Given the actions of private agents, the central bank conducts policy by saving directly in foreign assets (reserves), bypassing financial intermediation. We characterize the Ramsey optimal reserve



(a) IIP to GDP (excluding reserves)

(b) Reserves to GDP

Notes: A sample of 20 largest (by 2022 GDP) persistent deficit emerging economies, i.e., those which are net external debtors (excluding reserves), is considered.
The left panel shows the mean and median international investment position (IIP) of this sample, and the right panel shows the official foreign exchange reserves. The shaded region represents the maximum and minimum. In 2022, the median IIP was -44.86% and median reserves were 16.27% in this sample. Source: Computed using the External Wealth of Nations database (Milesi-Ferretti 2022).

Figure 1: Net-external debtors holding reserves

accumulation policy for an economy that enters the initial period with a stock of reserves on the central bank's balance sheet and an outstanding external debt position associated with past household borrowing through intermediaries. We show that by liquidating reserves, the central bank can feasibly implement the 'frictionless' allocation, one that eliminates intermediated household borrowing and achieves interest parity and therefore zero intermediation costs. However, given initial domestic-currency-linked debt, this allocation is never optimal.

When the central bank sells reserves, it allows the household to reduce reliance on external borrowing, thereby lowering the resource loss associated with this borrowing. However, as the demand for borrowing falls, domestic bond market clearing requires a decline in the real rate of return on these bonds, which necessitates a real exchange rate appreciation. Such an appreciation has a first-order pecuniary general equilibrium consequence: it induces a revaluation of pre-existing obligations, raising the real value of inherited debt in tradeable units, thereby increasing the resources transferred abroad.

The central bank therefore faces a trade-off: using reserves to reduce current and future intermediation costs comes at the expense of increasing the real value of inherited debt. The Ramsey optimal policy implements an allocation that minimizes the overall resource loss for the economy: it retains reserves, accepting some interest parity gaps and costly household borrowing to avoid an even more costly revaluation of inherited obligations. Moreover, the marginal revaluation cost of selling reserves is increasing in the tightness of financial intermediation conditions. As a result, the central bank retains more reserves and accepts larger interest parity gaps when private intermediation is disrupted.

We also analyze the time-consistency of the Ramsey optimal reserve accumulation policy and show that this policy requires commitment. This time-inconsistency stems from the forward-looking behavior of intermediaries, whose credit supply is increasing in the expected appreciation rate. A central bank that commits to a future exchange rate internalizes that it affects next period's real repayment burden through two channels: the ex post real return paid per unit of borrowing, as well as the quantity of debt intermediaries lend today in anticipation of future appreciation. A commitment plan therefore chooses a deliberately weaker future currency in order to limit the future repayment burden through both components and thereby reduce the overall resource loss from intermediation.

However, if the central bank is allowed to re-optimize when the future period arrives, the previously borrowed quantity now becomes a state variable and changing the exchange rate at that point affects the repayment burden only by changing the ex post real rate of return. The incentive to keep appreciation low is therefore weaker ex post than ex ante. Consequently, a discretionary policymaker faces an incentive to deviate toward a higher appreciation, i.e., a stronger currency than the commitment plan originally prescribed.

If a central bank lacks commitment and deviates to higher appreciation than promised, intermediaries anticipate this deviation and expand lending in the present. The central bank must then respond by accumulating even more reserves today to prevent an excessive appreciation driven by future expectations. We illustrate that given identical initial conditions, a time-consistent (Markov-perfect) equilibrium features greater reserve accumulation and a larger interest parity wedge in the initial period compared to the commitment equilibrium. In other words, lack of commitment can amplify the scale of reserve accumulation and interest parity deviation beyond what the commitment model generates.

Finally, we present a quantitative illustration of the time-consistent equilibrium with financial flow volatility driven by shocks to intermediary borrowing constraints. While volatile financial intermediation and imperfect risk-sharing generate a precautionary motive for accumulating reserves as insurance against financial disruptions, we show that revaluation effects constrain the efficiency of this insurance. During periods of financial disruption, when external intermediation tightens, there is an incentive to deploy reserves to limit the capital-outflow-led depreciation and smooth consumption. However, the appreciation induced by such a reserve deployment also revalues inherited debt. So, for every dollar of reserves deployed, a part of this insurance payout leaks out to foreign creditors in the form of a higher real payment on existing claims. Therefore, revaluation effects create an endogenous cost of reserve deployment, limiting the insurance benefits reserves can provide. Moreover, with a tightening of intermediation conditions, the exchange rate also becomes more sensitive to reserve operations: any given amount of reserve deployment induces a stronger appreciation, thereby raising the real value of inherited debt even more strongly.

We illustrate that the marginal revaluation cost of reserve deployment is increasing in the tightness of financial intermediation, making reserve utilization most costly precisely when tradeable resources are most scarce. A central bank that internalizes this state-dependent cost is therefore more restrained in deploying reserves. Our quantitative illustration shows that when revaluation costs are significant, the optimal policy may not be strictly procyclical to capital flows: the central bank finds it optimal to retain more reserves during disruptions than in normal times to avoid triggering costly debt revaluations.

The rest of this paper is organized as follows. Section 2 summarizes the related literature. Section 3 describes the model and establishes the competitive equilibrium conditions. Section 4 characterizes the optimal reserve policies and presents the main analytical results of this paper. Section 5 contains a quantitative analysis of the model. Section 6 concludes.

2 Related Literature

This paper is related to the literature on foreign exchange reserve accumulation motives. Existing work emphasizes mercantilist motives (Aizenman and Lee 2007), growth and capital accumulation (Korinek and Servén 2016; Benigno, Fornaro and Wolf 2022), and demand for safe assets (Caballero, Farhi and Gourinchas 2017). For net debtor economies holding both large reserves and costly external liabilities, the literature traditionally highlights precautionary and liquidity motives, where reserves provide self-insurance against sudden stops and capital-flow crises (Durdu, Mendoza and Terrones 2009; Obstfeld, Shambaugh and Taylor 2010; Jeanne and Rancière 2011; Gourinchas and Obstfeld 2012). Other recent explanations include sovereign rollover and default risk hedging

(Bianchi, Hatchondo and Martinez 2018; Bianchi and Sosa-Padilla 2024; Barbosa-Alves, Bianchi and Sosa-Padilla 2024), macroprudential policy addressing pecuniary and aggregate demand externalities (Arce, Bengui and Bianchi 2022; Bianchi and Lorenzoni 2022; Davis, Devereux and Yu 2023; Jeanne and Sandri 2023), and liquidity provision under currency mismatch (Bocola and Lorenzoni 2020). This paper takes the coexistence of reserves and external liabilities as given and asks why debtor economies do not net out their balance sheets to eliminate the carry cost associated with costly external liabilities and low-yielding reserves. We show that even in the absence of the above motives, liquidating reserves to pay down debt may not be optimal: selling reserves reduces demand for costly external borrowing but induces a real exchange rate appreciation that revalues inherited domestic-currency obligations, increasing their real burden. A central bank that internalizes the general equilibrium effects of its actions may not find it optimal to net out the country's balance sheet.

This paper is also related to the growing literature on financial intermediation frictions, and exchange rate management in the presence of such frictions. Traditionally, official reserve operations have been viewed as irrelevant for real allocations because private capital flows undo public balance-sheet changes, so only the net foreign asset position matters for equilibrium outcomes (Backus and Kehoe 1989). Empirically, many countries intervene in FX markets, consistent with 'fear of floating' and the prevalence of exchange rate controls (Calvo and Reinhart 2002; Ilzetzki, Reinhart and Rogoff 2019). Recent theoretical work shows how financial frictions break the benchmark irrelevance logic and make gross official positions consequential for exchange rates and allocations (Gabaix and Maggiori 2015). Complementing this, a growing empirical literature documents persistent deviations from covered interest parity conditions that strongly support balance-sheet constrained intermediation and segmented asset markets (Du, Tepper and Verdelhan 2018; Cerutti and Zhou 2023). Building on this line of work, this paper shows how using reserves to eliminate a costly intermediation wedge interacts with an endogenous revaluation of inherited exchange-rate-sensitive liabilities.

Our framework is also closely related to recent papers characterizing optimal exchange rate policies in the presence of frictions in the spirit of Gabaix and Maggiori (2015) and Itskhoki and Mukhin (2021). Cavallino (2019) derives optimal FX intervention rules and studies interactions with monetary policy. Amador, Bianchi, Bocola and Perri (2020) analyze interventions at the zero lower bound. Basu, Boz, Gopinath, Roch and Unsal (2020) characterize optimal policy mixes across multiple instruments. Fanelli and Straub (2021) show interventions can mitigate exchange rate volatility arising from a distributional pecuniary externality, though they are costly because foreign investors earn carry-trade profits. Itskhoki and Mukhin (2023) develop a general framework with nominal rigidities and intermediation frictions, showing optimal interventions eliminate risk-sharing wedges. While this paper belongs to the same class of models, namely a small open economy with limited arbitrage and intermediary balance sheet constraints, we focus on specific trade-offs faced by policy makers in debtor economies when the real value of debt in tradeable units is sensitive to real exchange rate movements. Our contribution is in analyzing how real-exchange-rate-induced revaluation effects influence optimal reserve and exchange rate management policies. Our framework shows that revaluation effects impose an endogenous constraint on how a central bank deploys reserves which has crucial implications for the economy's intertemporal consumption path.

Finally, this paper is related to the literature on time inconsistency of policy and ex-post expropriation via debt devaluation. Reserve operations in our environment move the real exchange rate and therefore the real value of domestic-currency-denominated payments to foreigners. The central bank can therefore use reserves to manipulate the exchange rate, and could in principle accumulate reserves today, then depreciate tomorrow to reduce the real burden of external payments. Therefore, this setup is naturally related to the classic debt devaluation literature in which governments have an incentive to use ex post inflation or depreciation to expropriate creditors (Calvo 1988; Bohn 1991; Fornaro 2022). However, our framework as well as equilibrium outcomes in our model are different from this literature in several important ways. First, the real structure matters: while classic expropriation models feature nominal rigidities or preset nominal contracts that governments devalue through unexpected inflation, our model is about reserve-induced movements in the real exchange rate and the resulting revaluation effects on liabilities denominated in units other than tradeables. Second, the instrument differs: reserve operations directly affect ex post returns of creditors rather than operating through monetary surprises. Third, in our framework, the incentive to devalue is constrained by consumption as well as intermediation costs. A real exchange rate depreciation that devalues existing debt also decreases current tradeable consumption of the household and induces intermediaries to lend more today, which raises the resource loss from costly intermediation. Therefore, in times when intermediary balance sheet constraints are more relaxed, the central bank finds it more costly to devalue and has a stronger incentive to liquidate reserves and appreciate.

The decisive difference, however, lies in the direction of ex-post deviation implied by our model. Ex-post expropriation models show that governments have an incentive to reduce the real value of payments through surprise inflation or depreciation relative to past promises. Using a three-period model, we show that, all else given, the direction of deviation relative to past promises in our framework is in fact opposite: once the debt is contracted, a discretionary planner next period would like to appreciate more than the commitment plan promised, allowing more contemporaneous consumption and a stronger currency, even though this raises the real value of existing debt payments. In other words, a discretionary planner ends up paying more than promised on inherited obligations and not less.

This opposite direction of deviation shows that time-inconsistency in our model operates through a dynamic credibility problem rather than strategic default or incentives to ex-post expropriate foreigners. We illustrate that the time-inconsistency is about the ex-post irrelevance of an ex-ante commitment device. Under commitment, the central bank internalizes that its future policy promise influences the future debt burden through two channels: directly, by determining the real rate of return paid tomorrow, and indirectly, by influencing the quantity borrowed today through intermediaries' forward-looking asset supply rule. The commitment policy is therefore restrained: promising a weaker future currency to limit future debt burden through both channels. However, when the future arrives and the central bank is allowed to re-optimize, the quantity of debt becomes a state variable. Adjusting the exchange rate at that point affects the debt burden only through the real rate of return, not through the quantity already borrowed. The incentive to restrain appreciation is therefore weaker ex-post than ex-ante. This structure parallels the classic time-inconsistency of capital taxation (Kydland and Prescott 1977; Fischer 1980; Lucas and Stokey 1983). Just as a government cannot credibly commit to low taxes after capital investment becomes sunk, the central bank cannot credibly commit to restrain appreciation after debt has been contracted. In both cases, the ex-ante policy promise serves as a disciplining device to influence private decisions, but becomes ex-post irrelevant once those decisions are made.

This paper has three main contributions. First, we identify and analyze the trade-off between eliminating a costly intermediation wedge and the endogenous loss associated with inherited debt revaluation. We characterize the Ramsey optimal policy in this setting and show that optimal policy retains reserves and accepts some intermediation costs to avoid costly revaluation of inherited obligations. Second, we demonstrate that this optimal reserve policy requires commitment. A central bank that lacks commitment has an incentive to deviate ex post to a stronger future appreciation. Consequently, the Markov-perfect equilibrium features greater reserve retention and larger interest-parity deviations than under commitment. Third, we show that tighter intermediation conditions are associated with greater reserve retention as the exchange rate is more sensitive to reserve operations in times of disruptions and, as a result, the central bank's marginal cost of deploying reserves is increasing in the tightness of intermediary borrowing constraints. We provide a quantitative illustration of the time-consistent equilibrium with shocks to intermediary balance sheet constraints, where this state-dependence becomes important: during disruptions, reserve sales that smooth consumption also raise real transfers to foreign creditors making reserve deployment endogenously costly. When revaluation costs are quantitatively important, optimal policy can feature greater reserve retention in disruption states than in normal states: reserves need not be procyclical with capital flows.

3 Model

This section presents a dynamic general equilibrium model of a pure exchange small open economy (SOE) with free capital mobility but facing financial intermediation frictions. The model consists of three types of agents, namely, a representative consumer, international financial intermediaries, and a central bank. Time is discrete and denoted by t. The agents are infinitely lived.

The following sub-sections describe the environment and the agents' optimization problems. The competitive equilibrium definition is stated, and the equilibrium conditions are characterized.

3.1 Representative Consumer

The representative consumer is infinitely lived and consumes two goods: an internationally tradeable consumption good: c_t, and a domestic non-tradeable good: m_t. Each period, the consumer receives a fixed endowment of the two goods: y and m^s, respectively. The consumer also receives a transfer, T_t, from the central bank.

The non-tradeable good, m, is the domestic unit of account and its price is normalized to 1 every period[1]. Let p_t denote the relative price of tradeable consumption, c_t, in units of the domestic

[1] m_t is a non-tradeable *good* with an endowment every period. It is the domestic unit of account: domestic prices and assets are denominated in units of this good

numeraire. The tradeable consumption good is assumed to be the world numeraire, its world price, $p^*_t = 1$. The law of one price holds in the tradeable goods market: $p_t = e_t p^*_t$ where e_t is the real exchange rate: price of tradeables in units of domestic non-tradeables.

In addition to consuming the two goods, the consumer also participates in a competitive domestic bonds market where it can save or borrow using one-period risk-free bonds denominated in units of the domestic numeraire: $\tilde{b}_{t+1}$. $\tilde{b}_{t+1} > 0$ implies saving whereas $\tilde{b}_{t+1} < 0$ implies borrowing from the domestic bonds market. The consumer is subject to a no-Ponzi-games constraint. Let R_{t+1} denote the gross risk-free interest rate in this market in units of the numeraire. Moreover, the bond markets are segmented: while the consumer can access the domestic bonds market, it does not have direct access to the international bonds market.

Let β denote the consumer's discount factor, σ be the risk aversion parameter, and ω be the relative utility preference parameter for the tradeable consumption good. For simplicity, it is assumed that the consumer's preferences are separable and homothetic in the two goods. The consumer's optimization problem is described by (1) and involves maximizing expected lifetime utility subject to the budget constraints (and a no-Ponzi-games constraint).

$$\begin{gathered}\max_{\{c_t, m_t, \tilde{b}_{t+1}\}} \quad \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left(\omega \frac{c_t^{1-\sigma}}{1-\sigma} + \frac{m_t^{1-\sigma}}{1-\sigma} \right) \qquad \text{s.t.} \\ e_t c_t + m_t + \tilde{b}_{t+1} = e_t y + R_t \tilde{b}_t + m^s + T_t \end{gathered} \tag{1}$$

The optimality conditions for (1) are given by an Euler equation, (2), an intra-temporal relation between c_t and m_t, (3) and a transversality condition, (4).

$$c_t{}^{-\sigma} = \beta R_{t+1} \mathbb{E}_t \left(\frac{e_t}{e_{t+1}} c_{t+1}{}^{-\sigma} \right) \tag{2}$$

$$m_t^{\sigma} = \frac{1}{\omega} e_t c_t{}^{\sigma} \tag{3}$$

$$\lim_{J \to \infty} \beta^J \mathbb{E}_t \left(c_{t+J}^{-\sigma} \frac{\tilde{b}_{t+J}}{e_{t+J}} \right) = 0 \quad \forall t \tag{4}$$

3.2 Financial Intermediaries

The world is populated by a unit mass of identical financial intermediaries who intermediate borrowing and lending transactions in the financial markets. The intermediaries are assumed to be owned outside the SOE, i.e., any profits or losses they make, accrue to the foreign owners of these intermediaries. The intermediaries have access to an international bonds market where they trade one-period risk-free bonds denominated in units of the world numeraire, i.e., the tradeable consumption good: q^*_{t+1}. Let R^* denote the gross interest rate in the international bonds market. This rate is assumed to be fixed. We assume that $R^* < \beta^{-1}$.

While intermediaries have access to the international bonds market, the SOE has free capital mobility, so they also have unrestricted access to the domestic bonds market. Let $\tilde{q}_{t+1}$ denote the asset position of the intermediaries in the domestic bonds market denominated in units of the domestic numeraire.

A positive asset position in either market denotes saving whereas a negative position denotes borrowing. The intermediaries take offsetting positions and face a balance sheet constraint, (5).

$$\frac{\tilde{q}_{t+1}}{e_t} + q^*_{t+1} = 0 \tag{5}$$

The intermediaries are risk neutral and seek to maximize their expected return, $\mathbb{E}_t \nu_{t+1}$ in units of tradeables, from their asset positions:

$$\mathbb{E}_t \nu_{t+1} = \mathbb{E}_t \left[\frac{e_t}{e_{t+1}} R_{t+1} - R^* \right] \frac{\tilde{q}_{t+1}}{e_t} \tag{6}$$

We now describe the key financial intermediation friction in this environment. Following Gabaix and Maggiori (2015), friction takes the form of a limited repayment commitment on the part of the intermediaries. Every period, immediately after taking their asset positions, an intermediary can divert a portion $\Gamma_t \left| \frac{\tilde{q}_{t+1}}{e_t} \right|$, with $\Gamma_t > 0$, of the position it intermediates: $\left| \frac{\tilde{q}_{t+1}}{e_t} \right|$. If the intermediary diverts the funds, the position is unwound and the lenders recover a portion $\left(1 - \Gamma_t \left| \frac{\tilde{q}_{t+1}}{e_t} \right| \right)$ of their claims $\left| \frac{\tilde{q}_{t+1}}{e_t} \right|$. Since the lenders anticipate the incentives of the intermediary to divert funds, the intermediary is subject to a credit constraint such that expected discounted returns from the intermediation business are weakly higher than the returns earned by diverting the funds. This constraint is described by (7)[2].

$$\frac{1}{R^*} \mathbb{E}_t \nu_{t+1} \geqslant \Gamma_t \left| \frac{\tilde{q}_{t+1}}{e_t} \right| \times \left| \frac{\tilde{q}_{t+1}}{e_t} \right| \tag{7}$$

The intermediary's constrained optimization problem is then given by (8).

$$\max_{\tilde{q}_{t+1}} \frac{1}{R^*} \mathbb{E}_t \left[\frac{e_t}{e_{t+1}} R_{t+1} - R^* \right] \frac{\tilde{q}_{t+1}}{e_t} \qquad \text{s.t. to (7)} \tag{8}$$

[2]For further discussion on such limited commitment constraints and their micro-foundations see Gabaix and Maggiori (2015) and the references therein.

Since the objective in (8) is linear in $\tilde{q}_{t+1}$ and the constraint (7) is convex in $\tilde{q}_{t+1}$, the constraint, (7), always binds at the optimum. The solution to (8) is then given by an asset supply rule, (9).

$$\frac{\tilde{q}_{t+1}}{e_t} = \frac{1}{\Gamma_t}\left[\frac{R_{t+1}}{R^*}\mathbb{E}_t\left(\frac{e_t}{e_{t+1}}\right) - 1\right] \tag{9}$$

First note that given Γ_t and the interest differential, $\frac{R^*}{R_{t+1}}$, the intermediary's asset position is a linearly increasing function of the expected appreciation rate, $\mathbb{E}_t\left(\frac{e_t}{e_{t+1}}\right)$, with $\tilde{q}_{t+1} = 0 \Leftrightarrow \mathbb{E}_t\left(\frac{e_t}{e_{t+1}}\right) = \frac{R^*}{R_{t+1}}$. Secondly, the key variable in this equation is Γ_t which is implicitly a measure of friction in the financial markets. $\Gamma_t = 0$ is the frictionless model where the solution to the optimization problem gives the uncovered interest parity condition[3]. For any finite value of $\Gamma_t > 0$ uncovered interest parity fails. In other words, $\Gamma_t > 0$ drives a wedge between the interest differential, $\frac{R^*}{R_{t+1}}$ and the expected appreciation rate, $\mathbb{E}_t\left(\frac{e_t}{e_{t+1}}\right)$. The supply of funds by the intermediaries is an increasing function of this wedge: given the interest differential, the higher the expected appreciation rate, higher the amount that intermediaries are willing to save in the domestic bonds market. Moreover, the supply is also a function of Γ_t: an increase in frictions is equivalent to financial disruptions where the intermediaries are willing to lend smaller amounts given any expected rate of return. As illustrated by Gabaix and Maggiori (2015), it turns out that the expected appreciation rate, $\mathbb{E}_t\left(\frac{e_t}{e_{t+1}}\right)$ is not only an essential equilibrium variable but is also the key variable that adjusts in response to shocks to Γ_t.

In what follows, we describe the competitive equilibrium and optimal policies in both deterministic and stochastic settings. In the deterministic setting, we assume that $\Gamma_t = \Gamma > 0, \forall t$, and in the stochastic setting we assume Γ_t follows an exogenous Markov switching regime.

3.3 Central Bank

Since the SOE is assumed to have free capital mobility, the central bank does not have access to any distortionary capital taxation instrument. It only has access to lump-sum transfers/taxes, T_t, denominated in units of the domestic numeraire. However, the central bank can bypass the intermediaries and directly access the international bonds market, where it can save in one-period risk-free foreign assets denominated in units of the tradeable consumption good, henceforth called

[3]If $\Gamma_t = 0$ intermediaries are simply a veil, the solution to their optimization problem gives the UIP condition: $\frac{R^*}{R_{t+1}} = \mathbb{E}_t\left(\frac{e_t}{e_{t+1}}\right)$

reserves: a_{t+1}. The central bank faces a period-by-period budget constraint, (10).

$$e_t a_{t+1} + T_t = e_t R^* a_t \tag{10}$$

We assume that while the central bank can save in foreign assets, borrowing directly through the international bonds market is not an option[4]: $a_{t+1} \geqslant 0$.

3.4 Competitive Equilibrium

Given $(a_0, \tilde{b}_0)$, central bank policy $\{T_t, a_{t+1}\}$, and an exogenous stochastic process for Γ_t, a competitive equilibrium is given by stochastic sequences of the real exchange rate, $\{e_t\}$, the interest rate on domestic bonds, $\{R_{t+1}\}$, consumption of the two goods, $\{c_t, m_t\}$, and asset positions in the domestic bonds market, $\{\tilde{b}_{t+1}, \tilde{q}_{t+1}\}$, so that,

- Given the exchange rates and interest rates, the representative consumer solves its optimization problem (1).
- Given the exchange rates and interest rates, the intermediaries solve their optimization problem, (8).
- The central bank's budget constraint, (10) holds every period.
- The domestic non-tradeable and the domestic bond markets clear every period:

$$m_t = m^s \tag{11}$$

$$\tilde{b}_{t+1} + \tilde{q}_{t+1} = 0 \tag{12}$$

Next, we proceed with deriving the competitive equilibrium conditions. Since the endowment for the domestic non-tradeable good, m^s, is assumed to be constant, for simplicity and without loss of generality, it is normalized: $m^s = 1$. Then combining (11) and (3), we obtain (13).

$$e_t = \omega c_t^{-\sigma} \tag{13}$$

Equation (13) establishes an explicit equilibrium relation between the real exchange rate, e_t, and the consumption of the tradeable good c_t. This equation is essentially equivalent to a constant-elasticity-

[4]This assumption is not necessary or required for the forthcoming optimal policy analysis in a deterministic setting. In a stochastic setting, this borrowing constraint becomes important and generates precautionary saving given market incompleteness.

of-demand rule and suggests that a real exchange rate depreciation implies that the consumer is willing to consume a smaller amount of the tradeable good.

Secondly, combining (13) with the consumer's Euler equation, (2), pins down the interest rate on domestic bonds in non-tradeable units:

$$R_{t+1} = \beta^{-1} \quad \forall t \tag{14}$$

Combining the consumer's budget constraint in (1) with the central bank budget constraint, (10), and market clearing condition, (11), yields a resource constraint for the tradeable consumption good, henceforth called the balance of payments (BoP) condition, (15).

$$e_t c_t + \tilde{b}_{t+1} + e_t a_{t+1} = e_t y + R_t \tilde{b}_t + e_t R^* a_t \tag{15}$$

Let $\frac{\tilde{b}_{t+1}}{e_t} \equiv b_{t+1}$ and $\frac{\tilde{q}_{t+1}}{e_t} \equiv q_{t+1}$ i.e., private agents' savings expressed in units of the tradeable consumption good. Using (12) and (13) the BoP condition can be expressed in units of tradeables, (16).

$$c_t - q_{t+1} + a_{t+1} = y + R^* a_t - R\left(\frac{c_t}{c_{t-1}}\right)^{\sigma} q_t \tag{16}$$

The constraint (16) suggests that the 'real' interest rate on external debt, in units of the tradeable good, is given by $R\left(\frac{c_t}{c_{t-1}}\right)^{\sigma}$ where $\left(\frac{c_t}{c_{t-1}}\right)^{\sigma}$ denotes the ex-post appreciation rate.

Secondly, using (13), the intermediaries' supply of assets, (9), can essentially be expressed as a distorted Euler equation, (17).

$$q_{t+1} = \frac{1}{\Gamma_t}\left[\frac{R}{R^*}\mathbb{E}_t\left(\frac{c_{t+1}}{c_t}\right)^{\sigma} - 1\right] \tag{17}$$

As discussed previously, the asset supply by intermediaries is an increasing function of the expected appreciation rate, $\mathbb{E}_t\left(\frac{c_{t+1}}{c_t}\right)^{\sigma}$.

Finally, the transversality condition can be expressed as below:

$$\lim_{J\to\infty} \beta^J \mathbb{E}_t(c_{t+J}^{-\sigma} q_{t+J}) = 0 \quad \forall t \tag{18}$$

Given (a_0, q_0, c_{-1}), given a reserve accumulation policy of the central bank $\{a_{t+1}\}$, given an exogenous stochastic process for Γ_t, and given a no-Ponzi-games condition, the competitive equilibrium is summarized by a system of two stochastic difference equations in $\{c_t, q_{t+1}\}$: (16), (17) together with a transversality condition, (18).

The competitive equilibrium conditions explicitly establish the dual role of the real exchange rate. First, the real exchange rate is an intratemporal price that influences the consumer's willingness to consume tradeable goods relative to non-tradeable goods as illustrated by (13). Second, because the denomination of domestic and foreign assets differs, the real exchange rate is also an asset price: it affects the ex ante real intertemporal price in the domestic bonds market as well as the ex post real value of financial flows between households and intermediaries as illustrated by (17) and (16). Crucially, unlike models of perfect capital mobility where the exchange rate is a passive price that must be implied by interest parity conditions, in this model, it is more active: the real exchange rate adjusts to clear the domestic bonds market. As highlighted by Gabaix and Maggiori (2015), this asset price role of the real exchange rate is a key feature of this class of models with imperfect capital mobility and this has important implications for optimal reserve management policies.

In Appendix A we analyze this asset price role of the exchange rate in our framework: how it adjusts to clear the domestic bonds markets and how it responds to shocks to intermediary borrowing constraints. We show that an increase in Γ_t, which can be interpreted as a tightening of financial intermediation conditions due to tighter intermediary balance sheet constraints, is associated with a real exchange rate depreciation whereas a decrease in Γ_t, representing relaxed balance sheet constraints and stronger capital inflows, is associated with a real exchange rate appreciation.

In the next section we proceed to set up the optimal policy problem and analyze how reserve operations by the central bank affect domestic bonds market asset positions by private agents and therefore the equilibrium real exchange rate.

4 Optimal Reserves

4.1 A Ramsey Problem

We now proceed to set up and characterize the Ramsey optimal reserve accumulation policy for the central bank. The central bank is benevolent and chooses its reserve accumulation policy, $\{a_{t+1}\}$, to maximize the expected lifetime utility of the representative consumer, subject to the competitive equilibrium conditions. The constrained optimal policy problem is given by (19) and the Ramsey equilibrium is given by the solution to this problem.

$$\max_{\{c_t, q_{t+1}, a_{t+1}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{c_t^{1-\sigma}}{1-\sigma} \quad \text{s.t.}$$

$$\begin{aligned} c_t - q_{t+1} + a_{t+1} &= y + R^* a_t - R \left(\frac{c_t}{c_{t-1}}\right)^{\sigma} q_t \\ q_{t+1} &= \frac{1}{\Gamma_t}\left[\frac{R}{R^*}\mathbb{E}_t \left(\frac{c_{t+1}}{c_t}\right)^{\sigma} - 1\right] \\ a_{t+1} &\geqslant 0 \end{aligned} \tag{19}$$

The balance of payments constraint, (16), and the intermediaries' supply of funds rule, (17), are both implementability conditions to this problem. In particular, at $t = 0$, the intermediary supply rule at $t = -1$ is not an implementability condition; the previously contracted debt in units of non-tradeable, $\frac{q_0}{c_{-1}^{\sigma}}$, is a state variable for the planner. Therefore, at $t = 0$, the planner is free to choose c_0 and internalizes that this affects the real, tradeable units, value of initial debt payments, $Rc_0^{\sigma} \frac{q_0}{c_{-1}^{\sigma}}$. The Ramsey equilibrium assumes that the planner commits to all future policy at $t = 0$ and does not re-optimize at any future date. As a result, this previous debt valuation asymmetry arises only at $t = 0$ and not at any future date. These time-0 valuation effects are crucial for answering the motivating question of this paper. To this end, it is helpful to analytically analyze a two-period version of this policy problem in a deterministic setting, which is considered next.

4.1.1 Two-Period Problem: Intermediation Wedge vs. Revaluation Effects

In this sub-section, we analyze a two-period deterministic version of the policy problem (19). Let $t = 1, 2$, with the initial state variables (a_1, q_1, c_0) given at $t = 1$. We focus on the contextually relevant case in which the economy enters period 1 with positive reserves on the central bank's balance sheet: $a_1 > 0$ and an outstanding external household debt position in non-tradeable units, associated with past borrowing through intermediaries: $\frac{q_1}{c_0^{\sigma}} > 0$. After substituting the intermediary asset supply rule into the BoP constraints, the policy problem is given by (20).

$$
\begin{aligned}
&\max_{\{c_1,c_2,a_2\}} \frac{c_1^{1-\sigma}}{1-\sigma} + \beta\frac{c_2^{1-\sigma}}{1-\sigma} \qquad \text{s.t.} \\
&c_1 - \frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{c_2}{c_1}\right)^{\sigma} - 1\right] + a_2 = y + R^* a_1 - R c_1^{\sigma}\left(\frac{q_1}{c_0^{\sigma}}\right) \\
&c_2 = y + R^* a_2 - R\left(\frac{c_2}{c_1}\right)^{\sigma}\frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{c_2}{c_1}\right)^{\sigma} - 1\right] \\
&a_2 \geqslant 0
\end{aligned}
\tag{20}
$$

The two BoP constraints in (20) can be combined to obtain an intertemporal resource constraint, (21) and the non-negativity constraint on a_2 is given by (22).

$$
c_1 + \frac{c_2}{R^*} + \underbrace{\frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{c_2}{c_1}\right)^{\sigma} - 1\right]^2}_{B} = y + \frac{y}{R^*} + R^* a_1 - R^* q_1 - \underbrace{R^*\left[\frac{R}{R^*}\left(\frac{c_1}{c_0}\right)^{\sigma} - 1\right] q_1}_{A} \tag{21}
$$

$$
a_2 = \frac{c_2 - y}{R^*} + \frac{R}{R^*}\left(\frac{c_2}{c_1}\right)^{\sigma}\frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{c_2}{c_1}\right)^{\sigma} - 1\right] \geqslant 0 \tag{22}
$$

The policy problem (20) together with (21) and (22) resembles a standard two-period consumption savings problem, however, the intertemporal resource constraint, (21), contains two peculiar expressions, labeled A and B, that represent resource costs arising from the financial intermediation structure in this environment.

First, the expression labeled A represents ex post intermediary profits, ν_1, associated with past household borrowing at $t = 0$. This term is obtained by decomposing the value of initial debt payment, $Rc_1^{\sigma}\left(\frac{q_1}{c_0^{\sigma}}\right)$, into two parts: the dollar amount owed by the intermediaries to their lenders, $R^* q_1$, and the ex post intermediary profits, ν_1, which is given by expression A. Since intermediaries are assumed to be owned outside the SOE, these profits represent a resource transfer to foreign creditors over and above the world interest rate cost. Notably, however, while q_1, c_0 are state variables for the planner at $t = 1$, c_1 is a choice variable. A real exchange rate depreciation at $t = 1$, i.e., a lower c_1 decreases the real, tradeable units, value of these profits and therefore the overall value of debt payments made by the country to the intermediaries. Analogously, a real exchange rate appreciation at $t = 1$, i.e., a higher c_1 increases the real value of debt payments. The choice of c_1 affects the value of these profits, and therefore, the effective real value of resources leaving the economy, through the ex-post appreciation rate, $\left(\frac{c_1}{c_0}\right)^{\sigma}$.

Second, the expression labeled B, $\frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{c_2}{c_1}\right)^{\sigma}-1\right]^2$, which is also equal to Γq_2^2, represents the present value of ex ante profits[5], $\frac{\nu_2}{R^*}$, associated with new intermediated external finance between period 1 and 2. It captures the premium paid by the economy for accessing international markets through the intermediaries at $t = 1$. The quadratic term signifies that only the magnitude matters, i.e., it does not matter whether the financial intermediation technology is used for borrowing or saving, but what matters is the extent to which it is used. The magnitude of this cost depends on the interest parity wedge, i.e., how much $\left(\frac{c_2}{c_1}\right)^{\sigma} \lessgtr \frac{R^*}{R}$ with no cost if and only if $q_2 = 0$ i.e., $\left(\frac{c_2}{c_1}\right)^{\sigma} = \frac{R^*}{R}$. Therefore, both expressions A and B represent resource costs that originate from the financial intermediation structure assumed in this environment. Crucially, the assumption that intermediaries are owned outside the SOE is necessary to generate these resource costs. If intermediaries were domestically owned, the profits they make from intermediation would represent a transfer between residents rather than an external resource transfer, in which case both expressions A and B would vanish from the resource constraint. Moreover, the exchange rate dependence of the ex post profits stems from the assumption that households contracted with intermediaries in domestic currency (non-tradeable units) rather than dollars (tradeable units). We discuss these alternative assumptions in sub-section 4.1.3.

The intertemporal resource constraint, (21), together with the non-negativity constraint, (22), represent the set of feasible competitive equilibria that can be reached by the central bank's choice of a_2. This constraint set highlights the fact that Ricardian equivalence fails in this environment and the central bank, by choosing a reserve position, is able to influence the intertemporal consumption profile of the economy. The central bank internalizes how its choice of reserves affects the overall resource loss for the economy.

Given this constraint, this paper's motivating question can be stated as: why does the central bank not use reserves to implement the allocation that sets expression B to zero? This intermediation loss is minimized, in fact, eliminated entirely when the intertemporal consumption path satisfies the standard frictionless Euler equation: $\left(\frac{c_2}{c_1}\right)^{\sigma} = \frac{R^*}{R}$. At this allocation, the interest parity condition holds, households neither borrow from nor save with intermediaries ($q_2 = 0$), and the implied net foreign asset position ensures there is no carry cost. If the central bank begins with sufficient reserves initially, implementing this allocation is feasible. We call this allocation the 'frictionless

[5]Since the problem assumes commitment at $t = 1$, ex post profits at $t = 2$ are equal to ex ante profits at $t = 1$.

allocation' because it replicates the outcome that would prevail in a world without intermediation frictions.

However, it is almost immediate that minimizing expression B (current intermediation costs) does not necessarily minimize total resource transfers to foreigners. The presence of ex post profits, expression A, means that the choice of c_1 affects the real value of inherited debt payments to foreign intermediaries.

Abstracting for the moment from the non-negativity constraint and assuming that the economy begins with no initial debt, then it is trivially the case that the frictionless allocation would be the optimal policy choice. Intuitively, if the economy has no initial local currency debt and the central bank can costlessly intermediate with foreigners for the economy then it would choose to eliminate any costly private intermediation and directly handle all external financial transactions for the economy.

In the presence of positive initial local-currency debt however, the frictionless allocation cannot be optimal anymore. By choosing a lower c_1 than the level prescribed by the frictionless allocation, the central bank can devalue existing debt and thereby save resources: a margin ignored by the frictionless allocation. At the same time, moving away from the frictionless allocation means that the central bank must allow some costly private intermediation. Thus the central bank faces a trade-off between the resource gain from devaluing inherited debt and the resource loss from increased intermediation costs. How far does the central bank deviate from the frictionless allocation? It continues to lower c_1 and accepts intermediation costs as long as the marginal resource gain from devaluing inherited debt payments exceeds the marginal resource loss from increased intermediation costs. The optimal policy equates the marginal cost of higher intermediation costs to the marginal resource benefit from further devaluation[6].

While the discussion above explains the trade-off mechanically through the resource constraint, it can be better interpreted from the perspective of general equilibrium effects of reserve operations. Suppose the central bank begins with sufficient reserves to implement the frictionless allocation. As the central bank liquidates reserves and transfers resources to the household, the household

[6]It is important to note that "debt devaluation" here is relative to the real value of debt implied by the frictionless allocation and should not be outright interpreted as the central bank ex-post expropriating foreign creditors through strategic depreciation. The two-period setup analyzed here takes the inherited debt q_1 and reserves a_1 as given state variables, with no information about what consumption level c_1 might have been promised in the past, which in turn depends on the intermediation costs in the past. To understand the direction of deviation relative to past promises, one needs to analyze a dynamic policy problem where the central bank optimizes over multiple periods and is given an option to re-optimize in the future. We analyze this dynamic problem in Section 4.2.1.

demand for external borrowing starts falling. However, given the intermediary asset supply rule, they decrease lending only if the real rate of return on domestic bonds falls. The domestic bonds market clearing therefore requires the appreciation rate, $(c_2/c_1)^\sigma$, to fall which necessitates a real exchange rate appreciation at $t = 1$, i.e., a rise in c_1. While this equilibrium price adjustment decreases the quantity intermediated in the bonds market, it also creates a pecuniary general equilibrium effect: the real, tradeable units, burden of inherited debt starts increasing. The central bank is not a price taking agent; it internalizes the general equilibrium consequences of its actions. Specifically, with positive inherited local currency debt, the marginal resource costs from debt revaluation exceed the marginal gains from eliminating intermediation costs entirely. Therefore, rather than fully consolidating the country's balance sheet, the optimal policy retains reserves and tolerates some intermediation costs, thereby minimizing the economy's total resource loss.

Diagrammatic Illustration It is convenient to convey the intuition behind the optimal policy through a diagrammatic illustration that resembles the canonical consumption-savings problem. We begin by characterizing the intertemporal resource constraint, (21), graphically in Figure 2. Panel (a) illustrates how expressions A and B affect the feasible consumption set. The constraint RC_1 ignores both terms, showing the standard linear constraint. Incorporating expression A yields RC_2, which tilts inward along the c_1-axis reflecting higher debt payments when c_1 rises. Finally, RC_3 incorporates expression B (intermediation costs), creating the distinctive shape where the feasible set shrinks as $\left(\frac{c_2}{c_1}\right)^\sigma$ moves away from $\frac{R^*}{R}$. Notably, RC_3 is tangent to RC_2 at the point where intermediation costs vanish: $\left(\frac{c_2}{c_1}\right)^\sigma = \frac{R^*}{R}$, or equivalently, $q_2 = 0$.

Each point on RC_3 represents a competitive equilibrium for some reserve choice a_2. Panel (b) shows the frontier of RC_3 and marks two useful points. First, point P_{a0} corresponds to $a_2 = 0$: the allocation where the central bank holds no reserves. Moving leftward from P_{a0} along the feasible frontier corresponds to choosing $a_2 > 0$. Second, since q_2 is endogenous in $\frac{c_2}{c_1}$, every point on RC_3 also corresponds to a unique implied value for q_2. The figure marks point P_{q0} where $q_2 = 0$, which corresponds to the frictionless allocation that eliminates intermediation costs. The points to the left of P_{q0} correspond to $\left(\frac{c_2}{c_1}\right)^\sigma > \frac{R}{R^*} \Leftrightarrow q_2 > 0$, where the economy is borrowing through intermediaries. The relative positions of points P_{a0} and P_{q0} depend on the initial state (a_1, q_1, c_0).

A movement along the feasible frontier, accomplished by altering a_2, illustrates the failure of Ricardian equivalence in this environment: the central bank is able to influence the equilibrium

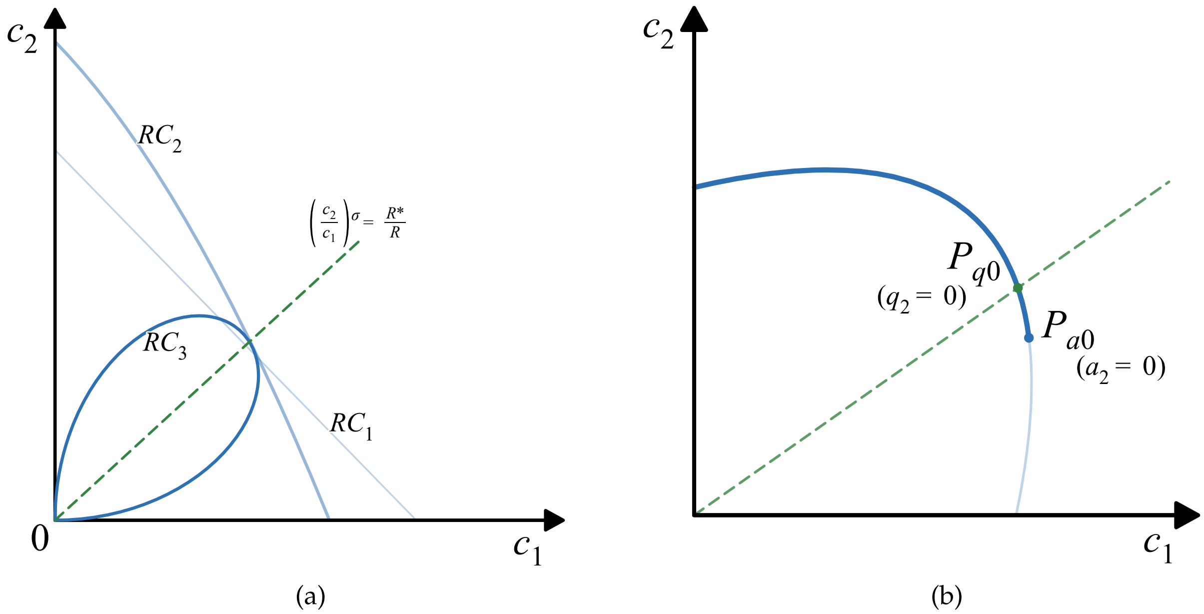
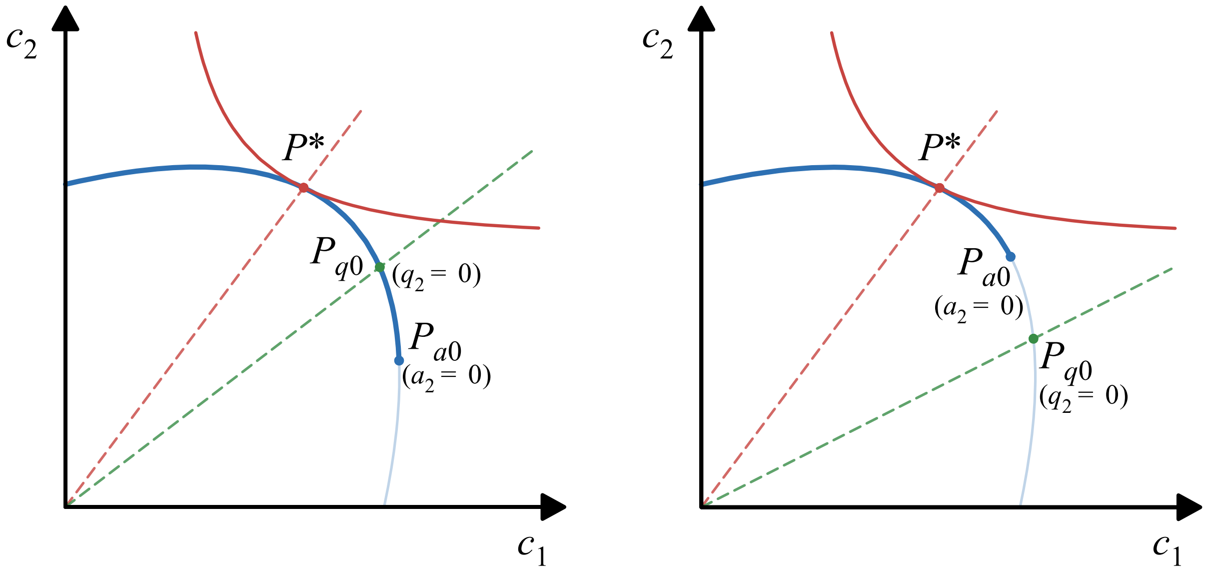


Figure 2: Intertemporal Resource Constraint

choices made by households and intermediaries and therefore the intertemporal consumption profile of the economy. In particular, Panel (b) shows that the central bank has sufficient initial reserves to feasibly implement the frictionless allocation at P_{q0} that eliminates the intermediation wedge. Consider increasing a_2 starting from P_{q0} by moving leftwards along the constraint. As the central bank accumulates reserves beyond this point, it induces the consumer to borrow more to offset this additional borrowing. However, the intermediaries are willing to lend more only if they are offered a higher appreciation rate, i.e., the exchange rate must depreciate at $t = 1$. Such a depreciation has three effects. First, it reduces period-1 consumption directly. Second, by increasing the appreciation rate, it induces more intermediary lending and raises intermediation costs. Third, it creates a general equilibrium effect that devalues inherited debt. These effects are captured in a leftward movement along the constraint away from P_{q0}. Figure 3 illustrates the optimal solution. Panel (a) shows the optimal allocation, P^*, where the indifference curve is tangent to the resource constraint. The optimal allocation is an interior solution that lies to the left of the frictionless allocation P_{q0}, highlighting the fact that at P_{q0} the marginal benefit of devaluing inherited debt exceeds the marginal cost of allowing the intermediation wedge. In other words, by moving from P_{q0} to P^*, the central bank is able to save more resources by devaluing inherited debt than the additional intermediation costs incurred. Any further devaluation is suboptimal as the intermediation costs would exceed the resource gain from devaluation. Conversely, any higher



(a) (b)

Figure 3: Optimal Policy

c_1 than the optimal solution is also suboptimal as the resource loss from debt revaluation would exceed the gain from reducing intermediation costs. The optimal solution therefore minimizes the overall resource loss to foreigners by balancing the marginal cost from revaluing inherited debt against the marginal benefit from private deleveraging.

Proposition 1 formally summarizes the main analytical result from this discussion. It shows that whenever the country begins with a positive stock of domestic-currency-linked external debt, eliminating the intermediation wedge is never optimal. In other words, whenever $\frac{q_1}{c_0^\sigma} > 0$, P_{q0} can never be optimal and P^* must lie to the left of P_{q0}: the marginal gain from devaluing inherited debt always exceeds the marginal cost of allowing some intermediation at P_{q0}. The planner decreases c_1 below the level prescribed by the frictionless allocation and tolerates some intermediation costs, thereby minimizing the overall resource loss to foreigners.

Proposition 1. *Given an initial state,* (c_0, a_1, q_1), *let the frictionless allocation be given by* $(\hat{c}_1, \hat{c}_2, \hat{q}_2, \hat{a}_2)$ *which satisfies* (21) *and* $\left(\frac{\hat{c}_2}{\hat{c}_1}\right)^\sigma = \frac{R^*}{R}$. *If* $\frac{q_1}{c_0^\sigma} > 0$, *then* $(\hat{c}_1, \hat{c}_2, \hat{q}_2, \hat{a}_2)$ *is not a solution to* (20). *Furthermore, the optimal solution involves:*

$$\left(\frac{c_2}{c_1}\right)^\sigma > \frac{R^*}{R}, \qquad c_1 < \hat{c}_1, \qquad c_2 > \hat{c}_2, \qquad q_2 > 0, \qquad a_2 > \hat{a}_2.$$

Proof. See Appendix B.1. □

So far we have ignored the non-negativity constraint on a_2 and the proof of Proposition 1 ignores this constraint. If the initial reserves are not large enough, it is possible that implementing the frictionless allocation is infeasible for the central bank. Then the most aggressive feasible way to minimize intermediation losses is to run reserves down to the lower bound $a_2 = 0$. However, the trade-off between intermediation costs and revaluation effects still applies. The optimal solution might still involve some positive reserves at $t = 1$[7]. Panel (b) of Figure 3 illustrates this case and Lemma 2 in Appendix B.2 establishes conditions under which this happens.

4.1.2 Comparative Statics with Respect to Γ

In this sub-section we present comparative statics with respect to the tightness of financial intermediation, Γ. As discussed previously, an increase in Γ is interpreted as a tightening of financial intermediation conditions due to tighter intermediary balance sheet constraints, whereas a decrease in Γ is interpreted as a relaxation of balance sheet constraints and stronger capital inflows. $\Gamma = 0$ corresponds to the frictionless world where gross asset positions are indeterminate and $\Gamma = \infty$ corresponds to financial autarky.

Mechanically, the constraint (21) shows that a higher Γ is associated with a smaller cost of allowing interest parity gaps (expression B is smaller), i.e., a smaller cost of deviating from the frictionless allocation. Therefore, as Γ increases, the central bank optimally chooses to deviate further from the frictionless allocation and allows a larger devaluation of inherited debt. In this sense, the intermediation costs impose an endogenous limit on the extent of devaluation that the central bank is willing to allow. As $\Gamma \to 0$, any deviation becomes increasingly costly and the optimal choice approaches the frictionless allocation. As $\Gamma \to \infty$, the cost of allowing deviations vanishes and the central bank is willing to allow an increasingly larger devaluation, however, the maximum devaluation is still limited by the fact that it comes at the cost of allowing lower consumption in period 1.

Intuitively, an increase in Γ implies that the marginal debt revaluation cost increases. When Γ is higher, reserve liquidation by a given amount induces a larger exchange rate appreciation. This is because when Γ is higher, the intermediary asset supply rule is steeper. When households try to deleverage by borrowing less from intermediaries, the intermediaries demand a stronger appreciation to be willing to lend less. This stronger appreciation means that the real value of

[7] It is also possible that the unconstrained optimal solution requires $a_2 < 0$, in which case the constrained solution is $a_2 = 0$.

inherited debt payments increases more, which makes the central bank more reluctant to liquidate reserves and allows for a larger devaluation of inherited debt. If $t = 1$ features financial autarky, the marginal cost of debt revaluation is also the highest, as every dollar of reserves spent increases consumption one-for-one without any offsetting borrowing from intermediaries thereby inducing a strong appreciation.

Therefore, as Γ increases, the optimal policy involves a larger devaluation of inherited debt and a larger interest parity wedge implemented by retaining more and more reserves. Proposition 2 formally characterizes the asymptotic behavior of optimal policy as $\Gamma \to 0$ and $\Gamma \to \infty$. As $\Gamma \to 0$, the optimal policy converges to the frictionless allocation. As $\Gamma \to \infty$, the optimal policy accepts larger interest parity gaps $\left(\frac{c_2}{c_1}\right)^\sigma - \frac{R^*}{R}$. At $\Gamma = \infty$, the central bank does not face any cost of allowing interest parity deviations, however the fact that further devaluation requires lower consumption in period 1 still constrains the maximum devaluation that the central bank is willing to allow. This maximum devaluation allocation features strictly lower consumption and strictly higher reserves in period 1 compared to the frictionless allocation. While global comparative statics with respect to Γ do not guarantee that reserves are monotone increasing in Γ for all parameter values, we illustrate using a numerical example that optimal reserves are indeed non-decreasing in Γ for reasonable parameter values. This is shown in Appendix C.1.

Proposition 2. *Given an initial state* (c_0, a_1, q_1) *and assuming* $\sigma \geqslant 1$, *let* $(c_1^*(\Gamma), c_2^*(\Gamma))$ *denote the unique interior optimizer of* (20) *for each* $\Gamma > 0$. *Then:*

- $x^*(\Gamma) \equiv \left(\frac{c_2^*(\Gamma)}{c_1^*(\Gamma)}\right)^\sigma$ *is strictly increasing in* Γ *and satisfies* $\lim_{\Gamma \downarrow 0} x^*(\Gamma) = \frac{R^*}{R}, \lim_{\Gamma \to \infty} x^*(\Gamma) = x^\infty$ *with* $\frac{R^*}{R} < x^\infty < \infty$.
- $c_2^*(\Gamma)$ *is strictly increasing in* Γ *and satisfies* $\lim_{\Gamma \downarrow 0} c_2^*(\Gamma) = \hat{c}_2, \lim_{\Gamma \to \infty} c_2^*(\Gamma) = c_2^\infty, c_2^\infty > \hat{c}_2$.
- $c_1^*(\Gamma)$ *admits limits:* $\lim_{\Gamma \downarrow 0} c_1^*(\Gamma) = \hat{c}_1, \lim_{\Gamma \to \infty} c_1^*(\Gamma) = c_1^\infty, c_1^\infty < \hat{c}_1$.
- $q_2(\Gamma) \to 0$ *as* $\Gamma \to \infty$, *and* $a_2(\Gamma) \to a_2^\infty \equiv \frac{c_2^\infty - y}{R^*}$, *with* $a_2^\infty > \hat{a}_2$.

Proof. See Appendix B.3. □

4.1.3 Alternative Assumptions

In this sub-section we consider modifying the set of assumptions that gives rise to the trade-off between eliminating the interest parity wedge and reducing losses due to revaluation effects. We show that in each of the cases discussed below, the trade-off disappears and the central bank optimally chooses to implement the frictionless allocation and eliminate any interest parity gaps.

No initial debt ($q_1 = 0$): If the country begins with no inherited debt, the expression A in (21) disappears. The central bank still faces a cost of deviating from the frictionless allocation given by expression B. As the central bank liquidates reserves, household borrowing decreases and the real exchange rate appreciates, however there are no revaluation costs. The overall resource loss of the economy also decreases as the economy moves towards the frictionless allocation. Therefore, in this case, the frictionless allocation is the optimal policy choice. If initial reserves are not sufficient to feasibly implement this allocation, it is still always optimal to run down reserves to the lower bound $a_2 = 0$ to minimize resource losses.

Frictionless world at $t = 1$ ($\Gamma = 0$): In the absence of the limited commitment friction at $t = 1$, the intermediation cost term in (21), expression B, vanishes. The intertemporal resource constraint simplifies to:

$$c_1 + \frac{c_2}{R^*} = y + \frac{y}{R^*} + R^* a_1 - R c_1^{\sigma} \frac{q_1}{c_0^{\sigma}}$$

Moreover, in this case, the intermediary asset supply is perfectly elastic, i.e., the interest parity condition holds, so that the intertemporal path of consumption satisfies the usual frictionless Euler equation: $\left(\frac{c_2}{c_1}\right)^{\sigma} = \frac{R^*}{R}$. The equilibrium is thus characterized by a unique net foreign asset position $(a_2 - q_2)$. In other words, any change in a_2 would be met by a one-for-one offsetting change in q_2 such that $(a_2 - q_2)$ does not change and hence no resultant change in c_1, c_2. Intuitively, due to a perfectly elastic supply of assets by the intermediaries, Ricardian equivalence holds: in response to the central bank accumulating reserves, the consumer borrows more, one-for-one so that the net foreign asset position does not change. This corresponds to the benchmark ineffectiveness of FX interventions result illustrated by Backus and Kehoe (1989). Therefore in this case, the trade-off between intermediation costs and debt revaluation through reserve operations disappears as the instrument itself is ineffective.

Domestically owned intermediaries: If intermediaries are domestically owned, any profits or losses made by intermediaries accrue to residents rather than foreigners. These profits are simply a transfer between residents rather than a resource loss for the economy. Notably, this applies to both the ex post profits associated with past borrowing (expression A) and the ex ante profits at $t = 1$, associated with new borrowing (expression B). In this case, the intertemporal resource constraint simplifies to the familiar linear constraint:

$$c_1 + \frac{c_2}{R^*} = y + \frac{y}{R^*} + R^* a_1 - R^* q_1$$

The optimal policy choice is again given by the frictionless Euler equation: $\left(\frac{c_2}{c_1}\right)^{\sigma} = \frac{R^*}{R}$.

This assumption implies that the debt between residents and foreigners is effectively denominated in tradeable units, so that the real value of debt payments is not affected by the real exchange rate. Therefore the central bank does not have the ability or incentive to devalue inherited debt.

Borrowing in tradeable units: If the consumer borrows through externally owned intermediaries in tradeable units, the real value of debt payments is not affected by the real exchange rate, i.e., the ex post real rate of return is also a state variable that does not depend on the real exchange rate at $t = 1$. Therefore, the central bank does not have the ability or incentive to devalue inherited debt through reserve operations. However, the intermediation resource loss associated with borrowing between period 1 and 2 still exists. Therefore the optimal policy is to eliminate this wedge and implement the allocation that satisfies the frictionless Euler equation.

In each of the cases discussed above, the trade-off between intermediation costs and revaluation effects disappears. The optimal portfolio at $t = 1$ involves netting out the country's balance sheet by running down reserves to eliminate the intermediation wedge ($q_2 = 0, a_2 > 0$) or if the initial reserves are insufficient to implement the frictionless allocation, then the optimal policy is to minimize resource losses by running down reserves to the lower bound ($a_2 = 0, q_2 > 0$). Therefore, the assumptions of positive domestic-currency-linked inherited debt, limited commitment intermediation friction at $t = 1$, and external ownership of intermediaries together generate this trade-off between eliminating the intermediation wedge and allowing initial debt revaluation that makes the frictionless allocation suboptimal. It is only in this case that optimal policy need not fully net out the country's balance sheet and close the interest parity gap ($q_2 > 0, a_2 > 0$).

4.1.4 Infinite Horizon

The results of the two-period problem discussed in section 4.1.1 can be generalized to the infinite horizon. Consider the deterministic version of the policy problem, (19) with $\Gamma_t = \Gamma, \forall t$. Suppose the economy begins in an initial state (q_0, a_0, c_{-1}) with $q_0 > 0$ and $a_0 > 0$. Analogous to the two-period problem, the BoP constraints can be combined to obtain an intertemporal resource constraint, (23). As before, the expressions A and B are labeled in (23). The expression A signifies the profits paid to intermediaries, over and above the interest cost on previously contracted debt. Due to debt being contracted in domestic currency terms, this expression is subject to revaluation effects through the real exchange rate chosen at $t = 0$. The expression B signifies the present discounted value of intermediation costs, i.e., ex ante intermediary profits on new external transactions from $t = 0$ onward. This expression is increasing in the size of interest parity deviations, and it is zero only when the parity condition holds each period.

$$\sum_{t=0}^{\infty} \frac{c_t}{R^{*t}} + \underbrace{\frac{1}{\Gamma} \sum_{t=0}^{\infty} \frac{1}{R^{*t}} \left[\frac{R}{R^*} \left(\frac{c_{t+1}}{c_t} \right)^{\sigma} - 1 \right]^2}_{B} = \sum_{t=0}^{\infty} \frac{y}{R^{*t}} + R^* a_0 - R^* q_0 - \underbrace{R^* \left[\frac{R}{R^*} \left(\frac{c_0}{c_{-1}} \right)^{\sigma} - 1 \right] q_0}_{A} \tag{23}$$

Equation (23) makes the extension immediate. A policy that reduces reliance on intermediated external finance going forward moves the economy towards smaller interest parity deviations and therefore reduces the resource loss for the economy. But doing so requires a general equilibrium adjustment of the real exchange rate at $t = 0$ that, when $q_0 > 0$, raises the real value of the inherited domestic-currency-linked repayments. The planner therefore faces the same trade-off as in the two-period model: lowering the intermediation cost on new flows comes at the cost of raising the real burden of inherited external payments.

However, in the infinite horizon problem this trade-off is relevant only in the short run. Since $\beta R^* < 1$, in the long run equilibrium, the economy converges to a steady state with a persistent intermediation wedge and reserves hit the borrowing limit (i.e., the non-negativity constraint binds after some t). In other words, $\beta R^* < 1$ creates relative impatience, and as one would expect, the central bank finds it optimal to eventually run down the reserves to the borrowing limit over time[8]. Moreover, once the central bank has run down its reserves, the transition dynamics to the long-run steady state are identical to the laissez-faire dynamics discussed in Appendix A.

The trade-off between current and future intermediation losses and revaluation effects associated with initial debt, however, is important for short-run dynamics. In Appendix C.2 we illustrate the long-run and short-run dynamics generated by optimal policy using a numerical illustration. We

[8] This is analogous to the standard infinite-horizon consumption-savings problem with a borrowing constraint and $R^* < \beta^{-1}$, where any initial wealth $a_0 > 0$ is driven down to the borrowing limit over time: $a_0 \geqslant a_1 \geqslant a_2 ... \geqslant -\bar{a}$.

show that, while in the long-run, the economy converges to a steady state with a persistent interest parity wedge and zero reserves, the short-run interest parity gaps are increasing in the level of initial debt q_0. In other words, in response to higher initial debt, the central bank accepts larger interest parity gaps in the short run to limit the revaluation costs which are increasing in the level of initial debt. We also compare the short-run dynamics of the Ramsey optimal policy against a naive carry-cost minimization policy that immediately runs down reserves to zero to minimize the intermediation losses in the short-run. We illustrate that this policy creates a larger overall resource loss due to debt revaluation and therefore lower welfare than the Ramsey optimal policy.

So far we have focused on the Ramsey optimal policy which requires the central bank to promise and commit to future policies. The following sub-section discusses how the intermediary asset supply rule creates time-consistency issues for the central bank. We analyze the implications of the central bank's inability to commit to future policies on equilibrium outcomes and welfare.

4.2 Time Inconsistency of Ramsey Optimal Reserve Policy

In this sub-section we continue with the analysis of the deterministic version of the Ramsey optimal reserve policy (19). As discussed earlier, the Ramsey optimal policy requires the central bank to promise and commit to future policies. This is because the forward looking intermediary asset supply rule (which is also a distorted Euler equation) is an implementability condition for the planner ex ante. However, if the planner re-optimizes ex post, this condition is no longer a constraint and any amount borrowed previously through the intermediaries becomes a state variable.

To better illustrate this time-inconsistency problem and understand the direction of a potential ex post deviation from a previous plan, we consider a three-period version of the policy problem.

4.2.1 Three-Period Policy Problem: Commitment vs. Deviation

Consider the three-period deterministic version of (19). Let $t = 0, 1, 2$ with the initial state (a_0, q_0, c_{-1}) given at $t = 0$. Let $\Gamma_t = \Gamma, \forall t$. As before, we assume that the economy begins with some initial debt and reserves: $a_0, q_0 > 0$. After substituting the asset supply equations into the BoP constraints, the time-0 policy problem, i.e., the problem under commitment, is given by (24).

$$
\begin{aligned}
&\max_{\{c_0,c_1,c_2,a_1,a_2\}} \frac{c_0^{1-\sigma}}{1-\sigma} + \beta\frac{c_1^{1-\sigma}}{1-\sigma} + \beta^2\frac{c_2^{1-\sigma}}{1-\sigma} \quad \text{s.t.} \\
&c_0 - \frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{c_1}{c_0}\right)^{\sigma} - 1\right] + a_1 = y + R^*a_0 - Rc_0^{\sigma}\frac{q_0}{c_{-1}^{\sigma}} \\
&c_1 - \frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{c_2}{c_1}\right)^{\sigma} - 1\right] + a_2 = y + R^*a_1 - R\left(\frac{c_1}{c_0}\right)^{\sigma}\frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{c_1}{c_0}\right)^{\sigma} - 1\right] \\
&c_2 = y + R^*a_2 - R\left(\frac{c_2}{c_1}\right)^{\sigma}\frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{c_2}{c_1}\right)^{\sigma} - 1\right] \\
&a_1, a_2 \geqslant 0
\end{aligned}
\tag{24}
$$

Consider now a potential deviation from the commitment plan at $t = 1$, i.e., the planner is allowed to re-optimize and choose (c_1, c_2, a_2, q_2). The problem faced by the time-1 planner is identical to the two-period problem discussed previously, (20). Comparing the $t = 1, 2$ BoP constraints of (24) with those of (20), the source of time-inconsistency can be seen: the time-1 planner takes q_1 as a state variable. In other words, while the period-0 asset supply equation $q_1 = \frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{c_1}{c_0}\right)^{\sigma} - 1\right]$ is an implementability constraint for the time-0 planner, it is not a constraint for the time-1 planner. The time-0 planner internalizes that the choice of c_1 affects the quantity of borrowing received from the intermediaries at $t = 0$ and therefore the debt burden at $t = 1$. On the other hand, since the time-1 planner takes q_1 as a given, from his perspective, the choice of c_1 does not affect the quantity of debt burden, q_1, at $t = 1$.

Let $(c_0^c, c_1^c, c_2^c, a_1^c, a_2^c, q_1^c, q_2^c)$ denote the solution to (24), i.e., the solution under 'commitment', as the planner is required to make this choice at $t = 0$ and then must commit to this solution at $t = 1$. Suppose further that the initial state (a_0, q_0, c_0) is such that $a_1^c, a_2^c > 0$.

Now consider a hypothetical situation, in which, after having made the choices at $t = 0$, the planner is allowed to re-optimize at $t = 1$, by solving (20), taking (a_1^c, q_1^c, c_0^c) as state variables. Let the solution to (20) under this 'deviation' situation be given by $(c_1^D, c_2^D, a_2^D, q_2^D)$. How does c_1^c compare to c_1^D? It turns out that if $a_1^c, a_2^c > 0$, then $c_1^D > c_1^c$.

To understand why the time-0 planner chooses lower consumption c_1^c compared to what the time-1 planner would prefer, consider how each planner views the cost of increasing c_1. From the time-0 planner's perspective, the debt burden in period 1 can be decomposed into two components: the quantity of debt q_1, and the real interest rate $R\left(\frac{c_1}{c_0}\right)^{\sigma}$ paid per unit of debt:

$$\underbrace{R\left(\frac{c_1}{c_0}\right)^{\sigma}}_{\text{price}} \underbrace{\frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{c_1}{c_0}\right)^{\sigma} - 1\right]}_{\text{quantity}}$$

When the time-0 planner increases c_1, it triggers a real exchange rate appreciation which affects the future debt burden at period 1 through two channels. First, there is a quantity effect: a higher appreciation rate between period 0 and period 1 makes lending more attractive to intermediaries, inducing them to supply more credit at period 0. This increases the quantity of debt q_1. Second, there is a price effect: higher consumption c_1 directly increases the real interest rate $R\left(\frac{c_1}{c_0}\right)^{\sigma}$ paid per unit of debt at period 1. Through both channels, the choice of c_1 raises the total debt repayment burden, $R\left(\frac{c_1}{c_0}\right)^{\sigma} q_1$, in period 1. Recognizing this, the time-0 planner makes a restrained choice, i.e., chooses a lower c_1 to limit the period-1 repayment burden through both channels, and therefore also the resource loss associated with intermediary profits.

By contrast, when the time-1 planner considers increasing c_1, the quantity q_1 is a state variable. Therefore, the quantity effect is no longer operative: changing c_1 at this stage cannot alter how much debt was supplied by intermediaries in the previous period. The time-1 planner faces only the price effect: increasing c_1 raises the real interest rate $R\left(\frac{c_1}{c_0}\right)^{\sigma}$ on the existing debt q_1. Since the time-1 planner does not face the quantity effect that constrained the time-0 planner, the marginal cost of increasing c_1 is lower. Consequently, the time-1 planner prefers higher consumption, i.e., a stronger ex post real exchange rate appreciation: $c_1^D > c_1^c$. This result is formally summarized in Proposition 3.

Proposition 3. *Let* $(c_0^c, c_1^c, c_2^c, a_1^c, a_2^c, q_1^c, q_2^c)$ *be the solution to* (24) *given* (a_0, q_0, c_{-1}). *Suppose* $q_1^c, a_1^c, a_2^c > 0$. *Then at time* $t = 1$, *given* (a_1^c, q_1^c, c_0^c), *let* $(c_1^D, c_2^D, a_2^D, q_2^D)$ *be the solution to the two period sub-problem* (20).

Since $a_1^c, a_2^c, q_1^c > 0$, *then* $c_1^D > c_1^c$, $a_2^D < a_2^c$, $c_2^D < c_2^c$ *and* $q_2^D < q_2^c$.

Proof. See Appendix B.4 □

The direction of deviation is noteworthy. Relative to past promises, the central bank prefers ex-post appreciation, i.e., a stronger currency, all else equal. This contrasts with expropriation models (Calvo 1988; Bohn 1991) where governments exploit surprise inflation or depreciation to reduce real debt burdens. Here, a discretionary planner pays more on inherited obligations than the commitment plan prescribed, not less. The time-inconsistency therefore reflects a credibility problem rather than incentives for ex-post expropriation. Once debt is contracted, the quantity effect vanishes and the central bank loses the restraint it exercised ex ante to contain borrowing. The commitment device that ex ante disciplines intermediary lending becomes ex post irrelevant for debt already incurred. This parallels the classic time-inconsistency of capital taxation: a government cannot commit to low taxes after investment is sunk (Kydland and Prescott 1977; Fischer 1980). Here, the central bank cannot commit to restrain appreciation after debt is contracted. In both cases, the policy promise guides private decisions ex ante but loses force ex post.

4.2.2 Infinite Horizon

This time-inconsistency result discussed above extends immediately to the infinite horizon policy problem. A time-0 planner announces future policies but then at each $t > 0$, the planner has an incentive to deviate from the announced plan and act like a time-0 planner. The time-inconsistency of Ramsey optimal policy is formally summarized in Proposition 4.

Proposition 4. *Let* $\{c_t, a_{t+1}, q_{t+1}\}_{t=0}^{\infty}$ *be the time-0 Ramsey plan that solves the deterministic version of* (19) *given* (a_0, q_0, c_{-1})*. Then at any* $T > 0$*, if* $a_T, a_{T+1}, q_T > 0$*, the continuation of the time-0 plan,* $\{c_t, q_{t+1}, a_{t+1}\}_{t=T}^{\infty}$*, is not a Ramsey plan that solves* (19) *given* (a_T, q_T, c_{T-1})*.*

Figure 11 in Appendix C shows a numerical simulation illustrating Proposition 4.

Given the time-inconsistency of the Ramsey optimal reserve policy, in the next sub-section we characterize a time-consistent (Markov-perfect) equilibrium.

4.2.3 Time-Consistent Equilibrium under Lack of Commitment

A central bank that lacks commitment will deviate from announced plans in the future. Anticipating such deviations, market participants adjust their choices in the present. This sub-section analyzes the optimal time-consistent policy for the three-period case discussed above.

At $t = 1$, the policy problem is given by (20) taking (a_1, q_1, c_0) as state variables. Let $\mathcal{C}_1(a_1, q_1, c_0)$, $\mathcal{C}_2(a_1, q_1, c_0)$ and $\mathcal{A}_2(a_1, q_1, c_0)$ be the optimal decision rules that solve (20). At $t = 0$, the planner internalizes these decision rules to solve for (a_1, q_1, c_0). The time-0 policy problem is given by (25).

$$\max_{\{c_0,a_1,q_1\}} \frac{c_0^{1-\sigma}}{1-\sigma} + \beta \frac{C_1(c_0,a_1,q_1)^{1-\sigma}}{1-\sigma} + \beta^2 \frac{C_2(c_0,a_1,q_1)^{1-\sigma}}{1-\sigma} \quad \text{s.t.}$$

$$c_0 - q_1 + a_1 = y + R^* a_0 - R c_0^{\sigma} \frac{q_0}{c_{-1}^{\sigma}} \tag{25}$$

$$q_1 = \frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{C_1(c_0,a_1,q_1)}{c_0}\right)^{\sigma} - 1\right]$$

The period-0 asset supply equation in (25) suggests that intermediaries internalize that the central bank lacks commitment and take future policy, $C_1(.)$, as a given when lending at $t = 0$. Similarly, the planner's objective also takes future policies as given. Let $(c_0^l, c_1^l, c_2^l, a_1^l, a_2^l, q_1^l, q_2^l)$ be the solution to this problem given (a_0, q_0, c_{-1}). How does this solution compare to the 'commitment' solution to (24)? Specifically, how does a_1^c compare to a_1^l? How does q_1^c compare to q_1^l? We show that if $a_1^c, a_2^c, q_1^c > 0$ then $a_1^l > a_1^c$ and $q_1^l > q_1^c$.

To see this, we start with the commitment plan $(c_0^c, c_1^c, c_2^c, a_1^c, a_2^c, q_1^c, q_2^c)$, that solves (24), and analyze how an anticipation of central bank deviating at $t = 1$ would affect equilibrium outcomes at $t = 0$. First, note that the commitment plan is no longer feasible for (25) as it violates the asset supply equation constraint for this problem. This follows from Proposition 3 where we established that $c_1^D \equiv C_1(a_1^c, q_1^c, c_0^c) > c_1^c$, i.e., at $t = 1$, given (a_1^c, q_1^c, c_0^c), the central bank would like to deviate to a higher level of consumption by lowering reserves. At $t = 0$, the intermediaries will anticipate this deviation, i.e., anticipating the ex-post appreciation rate to be higher, they would want to lend more at $t = 0$. In other words, the asset supply equation constraint in (25) is violated:

$$q_1^c < \frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{C_1(c_0^c,a_1^c,q_1^c)}{c_0^c}\right)^{\sigma} - 1\right]$$

Therefore, the solution to (24) is no longer feasible for the time-consistent problem. Intuitively, this can be thought of as the domestic bonds market clearing condition getting violated with the supply of funds exceeding the demand, i.e., there is excess lending by the intermediaries at $t = 0$ in response to the anticipated deviation in the future. This result is formally summarized in Lemma 1.

Lemma 1. *Let* $(c_0^c, c_1^c, c_2^c, a_1^c, a_2^c, q_1^c, q_2^c)$ *be the solution to* (24) *given* (a_0, q_0, c_{-1}). *Let* $a_1^c, a_2^c, q_1^c > 0$. *Then* (c_0^c, a_1^c, q_1^c) *is not a solution to* (25) *given* (a_0, q_0, c_{-1}).

Proof. See Appendix B.5 □

Given that the intermediaries anticipate the central bank to deviate at $t = 1$, they are willing to lend more at $t = 0$. This creates a situation of excess supply of funds: intermediaries are willing to lend more than what the consumer is willing to borrow at the current exchange rate. To restore the equilibrium, the exchange rate must appreciate at $t = 0$, i.e., even if the central bank does not change its reserve position, a_1^c, c_0 will increase to absorb the excess lending and clear the bond market. However, such an exchange rate appreciation will also revalue previous debt, $R\left(\frac{c_0}{c_{-1}}\right)^{\sigma} q_0$. This creates a motive for the central bank to respond by increasing its reserve position and preventing excessive appreciation. In other words, the excess lending by the intermediaries is partially absorbed through an exchange rate appreciation and partially by the central bank increasing its reserves. Therefore in the time-consistent equilibrium, it is the case that $c_0^l > c_0^c$, $q_1^l > q_1^c$ and $a_1^l > a_1^c$. This result is formally summarized in Proposition 5.

Proposition 5. *Given* (a_0, q_0, c_{-1})*, let* $(c_0^c, c_1^c, c_2^c, a_1^c, a_2^c, q_1^c, q_2^c)$ *be the solution to* (24)*. Suppose* $a_1^c > 0, a_2^c, q_1^c > 0$*. Let* $(c_0^l, a_1^l, q_1^l, q_2^l)$ *solve* (25) *given* (a_0, q_0, c_{-1})*. Then* $a_1^l > a_1^c$*,* $q_1^l > q_1^c$*, and* $c_0^l > c_0^c$*.*

Figure 12 in Appendix C shows a numerical example comparing the equilibrium solutions under commitment and lack of commitment, thereby illustrating Proposition 5, in particular, $a_1^l > a_1^c$ and $q_1^l > q_1^c$.

Proposition 5 establishes that given identical initial conditions, an economy under lack of commitment accumulates more reserves and features a larger interest parity wedge compared to an economy following the commitment plan. In other words, lack of commitment can amplify both reserve holdings and interest parity gaps relative to the commitment equilibrium. If the central bank cannot commit to future policies, this affects the trade-off it faces today: greater reserve retention is needed to contain an excessive exchange rate appreciation and the resulting debt revaluation, but this comes at the cost of accepting larger interest parity deviations.

In the next section, we describe a time-consistent equilibrium for the infinite horizon policy problem. Furthermore, with an understanding of the analytical properties of the deterministic policy problem discussed thus far, we consider the stochastic policy problem and analyze how revaluation effects influence optimal time-consistent policy given financial flow volatility.

5 Time-Consistent Policy in a Stochastic Environment

Optimal policy discussion so far has been restricted to a deterministic setting while assuming that the central bank can commit to future policies. In this section, we relax both assumptions. We set up the policy problem recursively, define a Markov-perfect equilibrium, and use a test calibration to solve for state-contingent decision rules. We discuss how revaluation effects create an endogenous state-dependent cost of reserve deployment and how this influences optimal time-consistent policy.

5.1 A Time-Consistent Policy Problem

We assume that the central bank cannot commit to future policies and that the economy is subject to financial flow shocks. As shown in the previous section, lack of commitment implies that the central bank takes future policies as given when making decisions in the present. Therefore, it is useful to set up the policy problem recursively. For the sake of computational ease, it is also useful to set up the problem with the exchange rate, e, as an explicit control variable. Let the endogenous state variables be given by the reserve position at the beginning of a period, a, and the existing debt denominated in domestic currency, $\tilde{q}$. We assume that Γ is the only exogenous state variable and follows a Markov switching regime.

Let $V(.)$ denote the value function associated with the consumer's utility function and let $\mathcal{E}(.)$ denote the exchange rate policy function. The control variables are choices of reserves, a', domestic currency denominated borrowing, $\tilde{q}'$, and the exchange rate, e. The central bank's recursive policy problem is given by (26).

$$
\begin{gathered}
V\big(a, \tilde{q}, \Gamma\big) = \max_{a', \tilde{q}', e} \left\{ \omega^{\frac{1}{\sigma}} \frac{e^{1-\frac{1}{\sigma}}}{1-\sigma} + \beta \mathbb{E}_{\Gamma'|\Gamma} \Big[V\big(a', \tilde{q}', \Gamma'\big) \Big] \right\} \qquad \text{s.t.} \\
\omega^{\frac{1}{\sigma}} e^{1-\frac{1}{\sigma}} - \tilde{q}' + e a' = e y - R\tilde{q} + e R^{*} a \\
\tilde{q}' = \frac{1}{\Gamma} \left[\frac{R}{R^{*}} \mathbb{E}_{\Gamma'|\Gamma} \left(\frac{e^2}{\mathcal{E}(a', \tilde{q}', \Gamma')} \right) - e \right] \\
a' \geqslant 0
\end{gathered}
\tag{26}
$$

First, we have substituted out c in terms of e in the objective function using (13). The first constraint is the BoP condition, i.e., equation (15), where we have substituted out c in terms of e and $(\tilde{b}, \tilde{b}')$ in terms of $(\tilde{q}, \tilde{q}')$ using (12). The second constraint is the intermediaries' asset supply equation (9).

Notably, $\mathcal{E}(.)$ represents the future exchange rate policy, reflecting the assumption that the central bank lacks commitment and the market participants internalize this, i.e., they take future exchange rate policy as a given while making their lending decisions. The equilibrium is defined below.

5.1.1 Markov-Perfect Equilibrium

A time-consistent (Markov-Perfect) equilibrium is defined by the optimal exchange rate and borrowing decision rules $\left(\mathcal{E}(.), \tilde{Q}(.)\right)$, a reserve accumulation rule $\mathcal{A}(.)$, and a value function $V(.)$ such that:

- Given a conjectured future exchange rate policy, $\mathcal{E}(.)$, and a reserve accumulation rule $\mathcal{A}(.)$, $\left(\mathcal{E}(.), \tilde{Q}(.)\right)$ constitute a competitive equilibrium.
- Given conjectured future exchange rate policy, $\mathcal{E}(.)$, and the competitive equilibrium, $\left(\mathcal{E}(.), \tilde{Q}(.)\right)$, the central bank follows a reserve accumulation rule $\mathcal{A}(.)$ such that $V(.)$ attains a maximum.
- The conjectured future exchange rate policy is indeed correct: $\mathcal{E}(.)$.

5.2 Results

In this section we present a quantitative solution to the time-consistent equilibrium described above. We analyze the state contingent policy functions for reserves, borrowing, and exchange rates and illustrate how revaluation costs influence the central bank's reserve accumulation policy. We show that at high debt levels, revaluation costs limit the insurance benefits of reserves, and reserve positions may become countercyclical to capital flows.

Calibration: As the purpose is illustrative, we use a test calibration to solve (26), however, the qualitative patterns below are robust across alternative calibrations we explored. A period in the model corresponds to a year. The domestic interest rate is assumed to be 4% ($R = \beta^{-1} = 1.04$). Since we assume that $R^* < R$, the world interest rate is set at 2% ($R^* = 1.02$). The risk aversion parameter, $\sigma = 2$ and the tradeable good preference parameter, $\omega = 0.3$. The tradeable good endowment is normalized $y = 1$. We assume that Γ is the only source of uncertainty in the model and that it follows a Markov switching regime with three states: $\Gamma \in \{0.005, 0.05, 10\}$.

We solve this problem using value function iteration. It involves solving a fixed-point problem nested within a dynamic programming problem as the policy functions depend on the future exchange rate policy, which must be consistent with current policies. We solve this using a nested iteration approach where we iterate on a guessed exchange rate policy function and value function until convergence.

5.2.1 Reserves as Insurance

In this sub-section we present and analyze the state contingent policy functions for reserves, borrowing, and exchange rates. Notably, the problem resembles, in many ways, a standard consumption-savings problem with incomplete markets and borrowing constraints. Together, this set of assumptions generates a precautionary saving motive: the agent accumulates reserves when resources are abundant to smooth consumption in future disruption states.

In the context of our model, shocks to financial flows through Γ create a volatility in the net tradeable resources available to the economy. As discussed previously and shown in Appendix A, a lower value of Γ corresponds to periods of financial booms, a loosening of the credit constraint faced by the intermediaries, which results in increased net capital inflows. This results in an abundance of tradeable resources and a resulting real exchange rate appreciation that increases tradeable consumption. Conversely, a higher value of Γ corresponds to periods of financial disruptions, i.e., due to a tightening of the intermediation credit constraint the economy faces higher borrowing costs which results in net capital outflows accompanied by a real depreciation and a decline in consumption. Given that international risk sharing is incomplete, the volatility in tradeable consumption induced by financial cycles, therefore, creates a precautionary saving motive for the central bank: accumulate reserves when Γ is low and use them as a form of self-insurance to smooth consumption in states when Γ is higher.

Our quantitative solution confirms that this intuition holds in this model, at least partially. Figure 4 depicts a slice of the policy function for reserves, $a' = \mathcal{A}(a, \tilde{q}, \Gamma)$. The left panel shows the optimal reserve choice as a function of the beginning of period reserves, for a given level of debt. Similarly, the right panel shows this optimal choice as a function of initial debt, for a given level of initial reserves. In both panels, the policy functions for the lowest and the highest state of Γ are shown. The optimal time-consistent policy illustrates that when initial reserves are low, the central bank builds reserves ($a' \geqslant a$) in financially stable times, i.e., when Γ is low, and the economy receives higher financial inflows. This reserve stockpiling in financially stable times illustrates precautionary saving as a form of insurance against probable financial disruptions in future.

The figure shows that the precautionary saving motive is decreasing in the level of initial reserves. This is because, for any given level of initial reserves, a higher reserve accumulation today is also associated with a higher household borrowing. While such a portfolio potentially increases the access to resources in future disruption states, it also decreases the access to resources in future stable states (low Γ), and this loss is increasing in the level of initial reserves. At the same time, reserve accumulation in the low Γ state is increasing in the level of initial debt. This is because, for any given level of reserve accumulation, future consumption volatility is increasing in the level of initial debt. In other words, with higher initial debt levels, the planner increasingly accepts a loss in resources in future stable states (low Γ) in exchange for a larger insurance benefit in future disruption states (high Γ).

The Figure also shows that in states of financial disruption, when Γ is higher, tradeable resources are scarce, the precautionary saving motive disappears. In a counterfactual world where the central bank does not run down reserves, the economy would face a larger consumption drop. Therefore, the central bank has a strong incentive to use reserves to smooth consumption and mitigate the impact of financial disruptions ($a' \leqslant a$).

Figures 13 and 14 in Appendix C.4 depict policy function slices for borrowing and the real exchange rate, respectively. Figure 13 shows that borrowing is a decreasing function of the beginning of period reserves and increasing in the level of initial debt. Moreover, the economy accumulates debt

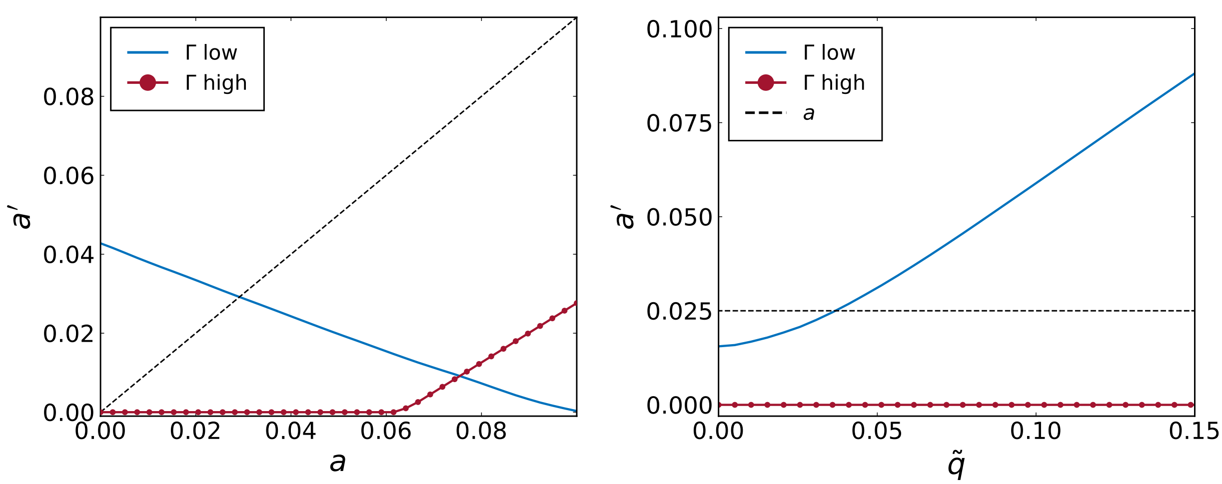


Figure 4: Policy Function- Reserves

through intermediaries in low Γ states and undergoes deleveraging in high Γ states. The exchange rate policy function illustrated in Figure 14 shows that starting a period with more reserves allows for a relatively appreciated exchange rate (and higher consumption), whereas higher existing debt is associated with a more depreciated exchange rate and hence lower consumption. Finally, with an increase in Γ, optimal policy allows some exchange rate depreciation and a fall in consumption. Notably, if reserves were not deployed in high Γ states, this depreciation would be larger.

5.2.2 The Debt Revaluation Cost of Reserve Deployment

Our discussion on the general equilibrium effects of reserve accumulation in the deterministic model highlighted that when the central bank runs down reserves, the resulting exchange rate appreciation revalues the inherited debt and creates an endogenous resource cost of deploying reserves. For every dollar of reserves deployed, only a fraction benefits the household; the remainder leaks to foreign creditors through higher real payments on existing debt. A central bank that internalizes these revaluation effects is, therefore, more restrained in deploying reserves.

Not only does this intuition carry over to the stochastic framework, but also the implications become starker. If reserves are interpreted as a form of insurance, then a reserve policy that deploys reserves in times of financial disruptions to limit the capital-outflow-led depreciation and consumption loss, faces an endogenous cost in the form of debt revaluation. Said differently, the insurance benefits of reserves spill over to foreign creditors through an exchange rate appreciation induced by this intervention: not only the households, but also their foreign creditors benefit from such an appreciation. In a laissez-faire world, a financial disruption would trigger a depreciation that reduces consumption but also devalues inherited debt obligations: a natural hedge that benefits the domestic economy at the expense of foreign creditors. By deploying reserves, the central bank trades off this beneficial depreciation against consumption smoothing. The central bank accepts higher real payments to foreigners in exchange for higher domestic consumption, with the optimal policy balancing this trade-off.

We show that in the presence of revaluation effects, reserve deployment is costly, and this cost is state-dependent: it is increasing in the tightness of financial intermediation, Γ. We illustrate this in Appendix C.4 using our numerical example. The marginal cost of deploying reserves can be measured in terms of the decrease in real payments to foreigners when the central bank marginally increases reserves at the optimum and induces a depreciation. We show that this cost is increasing in Γ. In other words, reserve deployment is most costly, precisely when tradeable resources are also most scarce.

Analogous to the intuition in the deterministic model that we discussed in Section 4.1.2, this increasing marginal cost with respect to Γ operates through the elasticity of the intermediaries' asset supply. During financial disruptions (higher Γ), intermediation is more constrained, rendering the supply of external financing highly inelastic. When the central bank deploys reserves, it effectively reduces the domestic economy's need to borrow. To clear the bonds market at this lower level of borrowing, the exchange rate must appreciate. Crucially, when intermediaries are more constrained, they demand a stronger exchange rate appreciation to accommodate this reduction in lending. Consequently, a given deployment of reserves induces a larger exchange rate appreciation when Γ is higher, which in turn increases the real value of initial debt even more strongly. Financial disruptions and capital outflow episodes induce depreciations and lower tradeable consumption, which creates a motive to deploy reserves and smooth consumption. However, in these times, the exchange rate is also more responsive to reserve operations: a dollar of reserves deployed induces a larger appreciation, which increases the real value of inherited debt more strongly, i.e., the marginal leakage to foreign creditors is also larger.

The central bank internalizes this increasing marginal revaluation cost and is therefore more restrained in deploying reserves in disruption states. When revaluation costs are significant, the optimal policy chooses to deploy reserves less aggressively during disruptions than during normal times. In other words, the reserve policy function can be non-monotone in Γ. This pattern is illustrated in the left panel of Figure 4 in the region where $\alpha'(\Gamma_{\text{high}}) > \alpha'(\Gamma_{\text{low}})$, which reverses the standard precautionary ordering established in the previous subsection, where the low Γ state involved more reserve accumulation than the high Γ state. This pattern is also illustrated in Figure 5 which shows the slices of reserve policy functions for the mid and high states of Γ for a given level of initial debt and reserves respectively. Both panels of this figure show that the optimal reserve deployment is weakly increasing in Γ, i.e., the central bank is more restrained in deploying reserves during disruption states than relatively stable states. When the non-negativity constraint on reserves binds, the revaluation effects, while still present, are not a binding margin. However, in states where this constraint does not bind, revaluation costs are a binding margin on how aggressively the central bank utilizes reserves. In a higher Γ state, the central bank is more restrained in deploying reserves because the marginal appreciation induced by reserve deployment is also larger, which

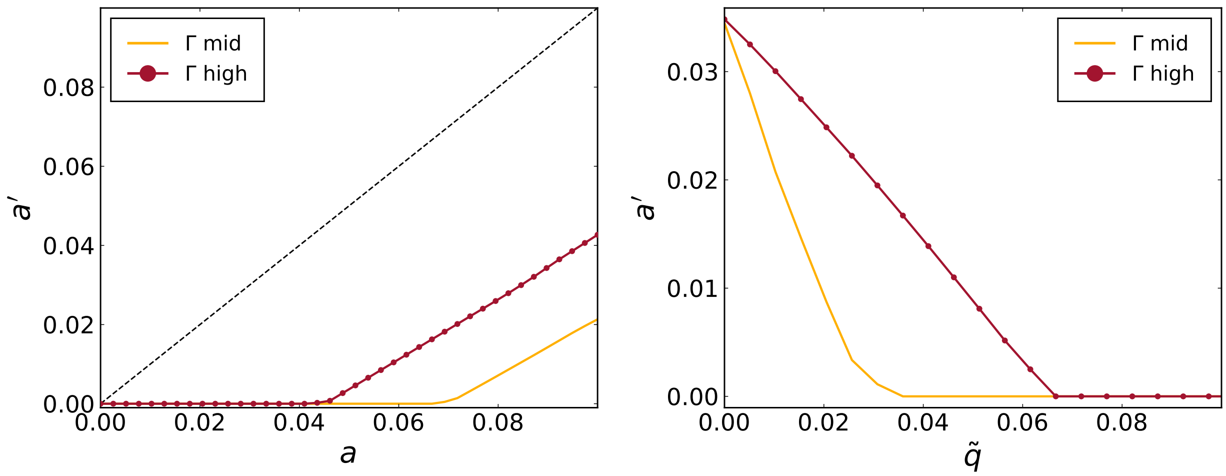


Figure 5: Policy Function- Reserve Deployment with high Revaluation Costs

increases the real value of initial debt more strongly.

Therefore, debt revaluation effects can create a non-monotonic relationship between reserves and capital flows. When debt is linked to domestic currency, deploying reserves during capital outflow episodes limits depreciation to support consumption, but this simultaneously revalues existing debt. When revaluation costs are large enough, they incentivize the central bank to retain more reserves rather than deploying them aggressively. Policy accepts more depreciation and lower consumption during financial disruptions to avoid a large exchange rate appreciation that would increase the real value of inherited claims. In simpler terms, revaluation effects associated with domestic-currency-linked external claims limit the insurance benefits that reserves can provide.

6 Conclusion

This paper studies why debtor emerging economies facing costly intermediation do not simply deploy reserves to eliminate the carry cost associated with external liabilities. We analyze optimal reserve and exchange rate policies in a small open economy with financial intermediation frictions and inherited local-currency debt. We show that while deploying reserves and appreciating the real exchange rate can eliminate costly intermediation, doing so triggers a general equilibrium revaluation effect that increases the real, tradeable units, burden of existing obligations. Consequently, the optimal policy, instead of netting out these opposing gross positions, retains reserves, accepting a persistent carry cost to avoid larger revaluation losses.

Moreover, the optimal reserve accumulation policy is time-inconsistent. The central bank has an incentive to announce a weaker future currency to mitigate the future debt burden ex ante, but ex post finds this policy sub-optimal. Instead, a discretionary policymaker prefers stronger ex post appreciation reflecting a dynamic credibility problem. In a time-consistent equilibrium, the central bank retains even more reserves and accepts larger interest parity gaps than under commitment.

Finally, we show that given an initial debt level, reserve accumulation is increasing in the tightness of financial intermediation. In a quantitative illustration of the time-consistent equilibrium with shocks to intermediary balance sheet constraints, we illustrate that debt revaluation costs are increasing in the tightness of intermediation, creating an endogenous cost of reserve deployment. When revaluation costs are significant, optimal policy may retain more reserves during disruptions than in normal times.

The paper abstracts from several important considerations. It does not compare policy instruments or analyze their interactions, such as the relative merits of capital controls versus reserves. Future work could extend the framework to incorporate nominal rigidities and monetary policy interactions and empirically examine how the currency composition of external liabilities shapes reserve deployment policies.

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Appendix

A Equilibrium Exchange Rate Determination and Dynamics

To better illustrate the determination of the equilibrium exchange rate and competitive equilibrium dynamics, this section describes the solution of a laissez-faire equilibrium in a deterministic setting. In other words, we assume that there is no uncertainty: $\Gamma_t = \Gamma > 0, \forall t$ and that the central bank follows a free-floating exchange rate regime so that it does not accumulate or hold reserves: $a_{t+1} = 0, \forall t$. Using deterministic versions of (16) and (17), a laissez-faire equilibrium is summarized by a system of two difference equations in $\{c_t, q_{t+1}\}$ (27), (28), and a transversality condition, (29).

$$c_t - q_{t+1} = y - R\left(\frac{c_t}{c_{t-1}}\right)^{\sigma} q_t \tag{27}$$

$$q_{t+1} = \frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{c_{t+1}}{c_t}\right)^{\sigma} - 1\right] \tag{28}$$

$$\lim_{t\to\infty} \beta^t (c_t^{-\sigma} q_{t+1}) = 0 \tag{29}$$

The steady state of this system is given by $(\bar{c}, \bar{q})$:

$$\bar{c} = y - (R-1)\bar{q}; \qquad \bar{q} = \frac{1}{\Gamma}\left[\frac{R}{R^*} - 1\right]$$

Since it is the case that $R^* < \beta^{-1} = R$ and $\Gamma > 0$, then $\bar{q} > 0$, i.e., the SOE carries a positive debt in the long-run equilibrium. To understand the dynamics of debt and consumption away from the long-run equilibrium, we present a phase diagram in the (q, c) space. Let $\Delta q = 0$ represent the zero-change locus for q and $\Delta c = 0$ represent the zero-change locus for c. Using (27) and (28) the loci can be expressed as below:

$$\Delta q = 0: \quad c = y + q\left(1 - R^*(1 + \Gamma q)\right)$$
$$\Delta c = 0: \quad q = \frac{1}{\Gamma}\left[\frac{R}{R^*} - 1\right]$$

The left panel of Figure 6 presents the phase diagram depicting these zero-change loci. The intersection of these loci signifies the long-run equilibrium $E_0 = (\bar{q}, \bar{c})$. The top-right quadrant has trajectories where consumption and debt explode. Such paths cannot constitute an equilibrium as they violate the no-Ponzi-games constraint. Similarly, the trajectories in the bottom-left quadrant cannot constitute an equilibrium as they involve the household accumulating wealth without

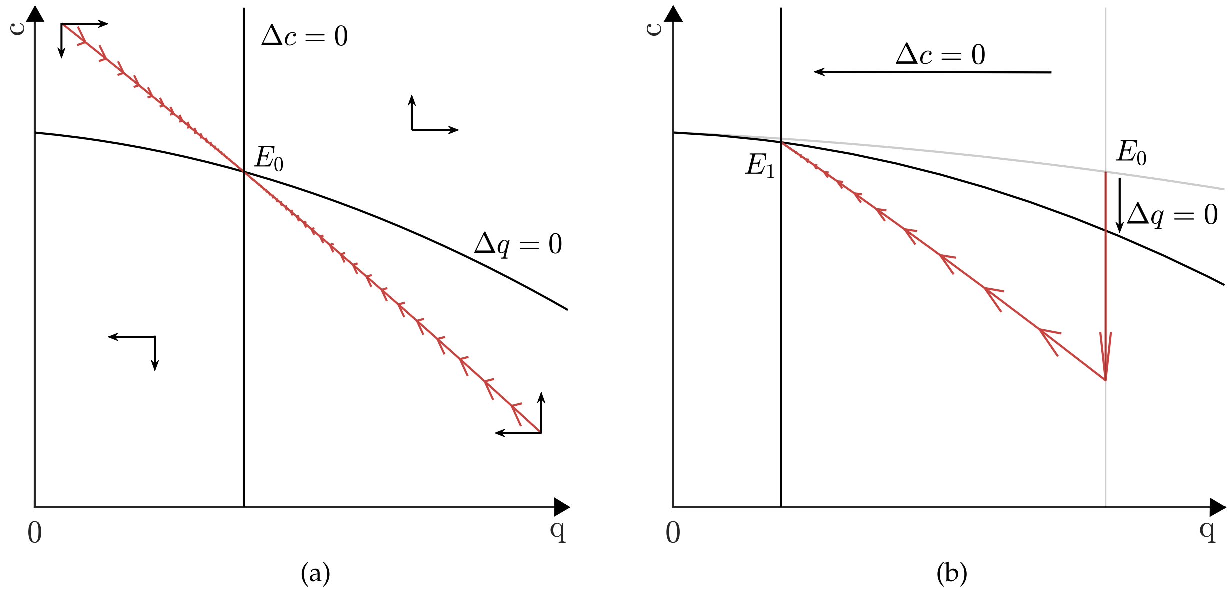

Figure 6: Phase diagram analysis

consuming anything asymptotically, thereby violating the transversality condition. However, the top-left and the bottom-right quadrants have a unique saddle path that asymptotically converges to the long-run equilibrium. If the household begins with low levels of debt (or high saving) it consumes more in the initial periods by borrowing and builds debt over time. As debt starts building up, higher debt payments cause consumption to decrease and the economy approaches the long-run equilibrium. Conversely, if the household starts with very high levels of debt, it initially lowers consumption to alleviate its debt burden. As debt decreases over time, the household can gradually increase its consumption, eventually approaching the long-run equilibrium.

Next, we demonstrate the adjustment and dynamics of the equilibrium exchange rate in response to unanticipated financial flow shocks. The right panel of Figure 6 illustrates the short-run and long-run impact of an unanticipated financial shock: a permanent increase in Γ. This increase is interpreted as a tightening of the credit constraint faced by the intermediaries, constituting a financial disruption. The shock induces a shift in both the loci and the new long-run equilibrium shifts from E_0 to E_1. From (13), noting the inverse relation between consumption and the exchange rate, it is observed that on impact, a financial disruption induces an exchange rate depreciation and a credit contraction. In the long run, the exchange rate appreciates towards its new long-run equilibrium. Analogously, a decrease in Γ, interpreted as a loosening of the credit constraint faced by the intermediaries, a financial boom, induces an exchange rate appreciation and a surge in capital inflows.

This analysis highlights the asset price role of exchange rates in models with imperfect financial markets: the exchange rate adjusts to ensure that the financial markets clear. Furthermore, periods of financial booms and busts are accompanied by corresponding exchange rate appreciations or depreciations in response to adjustments in financial flows.

B Proofs

B.1 Proof of Proposition 1

Proof. Let $q \equiv q_1/c_0^\sigma > 0$ and define:

$$x \equiv \left(\frac{c_2}{c_1}\right)^\sigma > 0, \qquad z \equiv \frac{R}{R^*}x - 1$$

Rewrite (21) as:

$$F(c_1, c_2) \equiv c_1 + \frac{c_2}{R^*} + \frac{1}{\Gamma}z^2 + Rq\, c_1^\sigma - W = 0, \qquad W \equiv y + \frac{y}{R^*} + R^* a_1 \tag{30}$$

Step 1: The optimal allocation satisfies $x > R^*/R$.

At an interior optimum, the tangency condition between the objective and the constraint is:

$$R\left(\frac{c_2}{c_1}\right)^\sigma = Rx = \frac{F_{c_1}}{F_{c_2}} \tag{31}$$

Differentiation of F yields

$$F_{c_1} = 1 + Rq\,\sigma c_1^{\sigma-1} - \frac{2\sigma}{\Gamma}\frac{R}{R^*}z\frac{x}{c_1} \qquad F_{c_2} = \frac{1}{R^*} + \frac{2\sigma}{\Gamma}\frac{R}{R^*}z\frac{x}{c_2}$$

Suppose, by contradiction, that $x \leqslant R^*/R$, equivalently $z \leqslant 0$. Then the last term in F_{c_1} is nonnegative, so

$$F_{c_1} \geqslant 1 + Rq\,\sigma c_1^{\sigma-1} > 1$$

and the last term in F_{c_2} is non-positive, so $F_{c_2} \leqslant 1/R^*$. Hence

$$\frac{F_{c_1}}{F_{c_2}} > R^*$$

But $x \leqslant R^*/R$ implies $Rx \leqslant R^*$. This contradicts (31). Therefore the optimum must satisfy

$$x = \left(\frac{c_2}{c_1}\right)^{\sigma} > \frac{R^*}{R} \tag{32}$$

Step 2: $(\hat{c}_1, \hat{c}_2)$ **cannot be optimal.**

By construction, the frictionless allocation satisfies $x(\hat{c}_1, \hat{c}_2) = R^*/R$. Step 1 shows any optimizer must satisfy $x > R^*/R$. Hence $(\hat{c}_1, \hat{c}_2)$ cannot solve (20).

Step 3: $q_2 > 0$**,** $c_1 < \hat{c}_1$**,** $c_2 > \hat{c}_2$**, and** $a_2 > \hat{a}_2$**.**

From Step 1, since $x > R^*/R$, we have $z > 0$. Then $q_2 = \frac{1}{\Gamma} z > 0$.

Let (c_1, c_2) denote the optimal solution. Suppose by contradiction that $c_1 \geqslant \hat{c}_1$. Using (32) and $\hat{x} = R^*/R$,

$$c_2 = c_1 x^{1/\sigma} \geqslant \hat{c}_1 x^{1/\sigma} > \hat{c}_1 \hat{x}^{1/\sigma} = \hat{c}_2$$

so $c_2 > \hat{c}_2$. Moreover, $z > 0$ while $\hat{z} = 0$. Hence, term by term in (30),

$$c_1 \geqslant \hat{c}_1, \qquad \frac{c_2}{R^*} > \frac{\hat{c}_2}{R^*}, \qquad \frac{1}{\Gamma} z^2 > 0 = \frac{1}{\Gamma}\hat{z}^2, \qquad Rq\, c_1^{\sigma} \geqslant Rq\, \hat{c}_1^{\sigma}$$

with at least one strict inequality. Therefore $F(c_1, c_2) > F(\hat{c}_1, \hat{c}_2) = 0$, contradicting feasibility $F(c_1, c_2) = 0$. Thus $c_1 < \hat{c}_1$.

Finally, since the objective is strictly increasing in both c_1 and c_2 and $(\hat{c}_1, \hat{c}_2)$ is feasible, the inequality $c_1 < \hat{c}_1$ rules out $c_2 \leqslant \hat{c}_2$ at the optimum; otherwise the objective at (c_1, c_2) would be strictly lower than at $(\hat{c}_1, \hat{c}_2)$, contradicting optimality. Hence $c_2 > \hat{c}_2$.

Finally, given that $a_2 = \frac{c_2 - y}{R^*} + \frac{R}{R^*} x\, q_2$ since $x > 0$ and $q_2 > \hat{q}_2 = 0$ and $c_2 > \hat{c}_2$, we have $a_2 > \hat{a}_2$. □

B.2 Lemma 2

Lemma 2. *Suppose that* $c_0 > 0, a_1 > 0, q_1 > 0$ *and suppose there exists* $(\hat{c}_1, \hat{c}_2, \hat{q}_2, \hat{a}_2)$ *that satisfies the constraints of* (20) *such that:*

$\hat{a}_2 = 0,$

$\hat{q}_2 = \frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{\hat{c}_2}{\hat{c}_1}\right)^{\sigma} - 1\right] \geqslant 0$

$\hat{c}_2 = y - R^*(1 + \Gamma \hat{q}_2)\hat{q}_2$

$\hat{c}_1 - \hat{q}_2 = y + R^* a_1 - R\left(\frac{\hat{c}_1}{c_0}\right)^{\sigma} q_1$

If

$$R^* a_1 + (\sigma-1)R\left(\frac{\hat{c}_1}{c_0}\right)^{\sigma} q_1 > \left[\frac{R}{R^*}\left(\frac{\hat{c}_2}{\hat{c}_1}\right)^{\sigma-1} - 1\right] y + \Gamma\hat{q}_2^2 + (2\sigma-1)(\hat{q}_2 + \Gamma\hat{q}_2^2)\left(R\left(\frac{\hat{c}_2}{\hat{c}_1}\right)^{\sigma-1} + 1\right)$$

then $(\hat{c}_1, \hat{c}_2, \hat{q}_2, \hat{a}_2)$ *is NOT a solution to* (20)*. Furthermore, the optimal solution involves* $a_2 > 0$ *and* $q_2 > \hat{q}_2$.

Proof. It suffices to establish the condition under which a perturbation to $(\hat{c}_1, \hat{c}_2, \hat{q}_2, \hat{a}_2)$ leads to a welfare gain. Consider a perturbation given by:

$q_2 = \hat{q}_2 + \Delta = \frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{c_2}{c_1}\right)^{\sigma}\right]$, where $\Delta > 0$

$c_2 = y + R^* a_2 - R^*(1 + \Gamma(\hat{q}_2 + \Delta))(\hat{q}_2 + \Delta)$

$c_1 + a_2 - (\hat{q}_2 + \Delta) = y + R^* a_1 - R\left(\frac{c_1}{c_0}\right)^{\sigma} q_1$

Differentiating wrt Δ:

$\frac{dc_1}{d\Delta} + \frac{da_2}{d\Delta} - 1 + R\sigma c_1^{\sigma-1}\frac{q_1}{c_0^{\sigma}}\frac{dc_1}{d\Delta} = 0$

$\frac{dc_2}{d\Delta} = R^*\left(\frac{da_2}{d\Delta} - (1 + 2\Gamma\Delta + 2\Gamma\hat{q}_2)\right)$

$1 = \sigma\frac{1}{\Gamma}\frac{R}{R^*}\left(\frac{c_2}{c_1}\right)^{\sigma}\left[\frac{1}{c_2}\frac{dc_2}{d\Delta} - \frac{1}{c_1}\frac{dc_1}{d\Delta}\right]$

Solving the above system of equations yields:

$$\frac{da_2}{d\Delta} = \frac{\frac{1}{\hat{\sigma}} + (1 + 2\Gamma\Delta + 2\Gamma\hat{q}_2)\frac{R^*}{c_2} + \frac{1}{c_1\kappa}}{\frac{R^*}{c_2} + \frac{1}{c_1\kappa}}$$

$$\frac{dc_2}{d\Delta} = R^*\left(\frac{\frac{1}{\hat{\sigma}} - \frac{1}{c_1\kappa}(2\Gamma\Delta + 2\Gamma\hat{q}_2)}{\frac{R^*}{c_2} + \frac{1}{c_1\kappa}}\right)$$

$$\frac{dc_1}{d\Delta} = -\frac{1}{\kappa}\left(\frac{\frac{1}{\hat{\sigma}} + (2\Gamma\Delta + 2\Gamma\hat{q}_2)\frac{R^*}{c_2}}{\frac{R^*}{c_2} + \frac{1}{c_1\kappa}}\right)$$

where $\kappa = 1 + \sigma R c_1^{\sigma-1}\frac{q_1}{c_0^{\sigma}}$ and $\hat{\sigma} = \sigma\left(\hat{q}_2 + \Delta + \frac{1}{\Gamma}\right) = \sigma\frac{1}{\Gamma}\frac{R}{R^*}\left(\frac{c_2}{c_1}\right)^{\sigma}$.

Since $q_1 > 0$, then $\hat{\kappa} = 1 + \sigma R\hat{c}_1^{\sigma-1}\frac{q_1}{c_0^{\sigma}} > 0$.

Since $\hat{q}_2 \geqslant 0$, then $\hat{\sigma} > 0$.

Let $\hat{D} \equiv \frac{R^*}{\hat{c}_2} + \frac{1}{\hat{c}_1\hat{\kappa}}$ where $\hat{D} > 0$.

Simplify further to obtain:

$$\left.\frac{dc_2}{d\Delta}\right|_{\Delta=0} = \frac{R^*}{\hat{\kappa}\hat{D}\hat{\sigma}}\left(\frac{\hat{c}_1\hat{\kappa} - 2\Gamma\hat{q}_2\hat{\sigma}}{\hat{c}_1}\right)$$

$$\left.\frac{dc_1}{d\Delta}\right|_{\Delta=0} = -\frac{1}{\hat{\kappa}\hat{D}\hat{\sigma}}\frac{\hat{c}_1}{\hat{c}_2}\left(\frac{\hat{c}_2 + R^* 2\Gamma\hat{q}_2\hat{\sigma}}{\hat{c}_1}\right)$$

Define welfare as:

$$W = \frac{\omega}{1-\sigma}\left(c_1^{1-\sigma} + \beta c_2^{1-\sigma}\right)$$

Change in welfare can be expressed as:

$$\frac{dW}{d\Delta} = \omega c_2^{-\sigma}\left(\left(\frac{c_2}{c_1}\right)^{\sigma}\frac{dc_1}{d\Delta} + \frac{1}{R}\frac{dc_2}{d\Delta}\right)$$

$$\begin{aligned}
\left.\frac{dW}{d\Delta}\right|_{\Delta=0} &= \omega \hat{c}_2^{-\sigma}\left(\left(\frac{\hat{c}_2}{\hat{c}_1}\right)^{\sigma}\left.\frac{dc_1}{d\Delta}\right|_{\Delta=0} + \frac{1}{R}\left.\frac{dc_2}{d\Delta}\right|_{\Delta=0}\right)\\
&= \frac{\omega \hat{c}_2^{-\sigma}}{\hat{\kappa}\hat{D}\hat{\sigma}}\frac{1}{\hat{c}_1}\frac{R^*}{R}\left(\hat{c}_1\hat{\kappa} - 2\Gamma\hat{q}_2\hat{\sigma} - \frac{R}{R^*}\left(\frac{\hat{c}_2}{\hat{c}_1}\right)^{\sigma-1}(\hat{c}_2 + R^* 2\Gamma\hat{q}_2\hat{\sigma})\right)
\end{aligned}$$

Then:

$$\begin{aligned}
&\left.\frac{dW}{d\Delta}\right|_{\Delta=0} > 0\\
\Leftrightarrow\quad & \left(\hat{c}_1\hat{\kappa} - 2\Gamma\hat{q}_2\hat{\sigma} - \frac{R}{R^*}\left(\frac{\hat{c}_2}{\hat{c}_1}\right)^{\sigma-1}(\hat{c}_2 + R^* 2\Gamma\hat{q}_2\hat{\sigma})\right) > 0\\
\Leftrightarrow\quad & \hat{c}_1 + \sigma R\left(\frac{\hat{c}_1}{c_0}\right)^{\sigma} q_1 - 2\sigma(\hat{q}_2 + \Gamma\hat{q}_2^2) > \frac{R}{R^*}\left(\frac{\hat{c}_2}{\hat{c}_1}\right)^{\sigma-1}(\hat{c}_2 + R^* 2\sigma(\hat{q}_2 + \Gamma\hat{q}_2^2))\\
\Leftrightarrow\quad & y + R^* a_1 + (\sigma-1)R\left(\frac{\hat{c}_1}{c_0}\right)^{\sigma} q_1 - 2\sigma(\hat{q}_2 + \Gamma\hat{q}_2^2) + \hat{q}_2 > \frac{R}{R^*}\left(\frac{\hat{c}_2}{\hat{c}_1}\right)^{\sigma-1}\left(y + (2\sigma-1)R^*(\hat{q}_2 + \Gamma\hat{q}_2^2)\right)\\
\Leftrightarrow\quad & R^* a_1 + (\sigma-1)R\left(\frac{\hat{c}_1}{c_0}\right)^{\sigma} q_1 > \left[\frac{R}{R^*}\left(\frac{\hat{c}_2}{\hat{c}_1}\right)^{\sigma-1} - 1\right] y + \Gamma\hat{q}_2^2 + (2\sigma-1)(\hat{q}_2 + \Gamma\hat{q}_2^2)\left(R\left(\frac{\hat{c}_2}{\hat{c}_1}\right)^{\sigma-1} + 1\right)
\end{aligned}$$

□

B.3 Proof of Proposition 2

Proof. Fix $q \equiv q_1/c_0^{\sigma} > 0$ and $\sigma \geqslant 1$. For each $\Gamma > 0$ let $(c_1^*(\Gamma), c_2^*(\Gamma))$ be the unique interior optimizer, and define

$$x^*(\Gamma) \equiv \left(\frac{c_2^*(\Gamma)}{c_1^*(\Gamma)}\right)^{\sigma}, \qquad z^*(\Gamma) \equiv \frac{R}{R^*}x^*(\Gamma) - 1, \qquad q_2(\Gamma) \equiv \frac{z^*(\Gamma)}{\Gamma}$$

Write the intertemporal resource constraint as $F(c_1, c_2; \Gamma) \leqslant W$, where

$$F(c_1, c_2; \Gamma) \equiv c_1 + \frac{c_2}{R^*} + \frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{c_2}{c_1}\right)^{\sigma} - 1\right]^2 + Rq\, c_1^{\sigma}, \qquad W \equiv y + \frac{y}{R^*} + R^* a_1$$

which binds at the optimum since utility is strictly increasing in (c_1, c_2).

Let $(\hat{c}_1, \hat{c}_2)$ as the frictionless allocation satisfying

$$\hat{c}_2 = \left(\frac{R^*}{R}\right)^{1/\sigma} \hat{c}_1, \qquad \hat{c}_1 + \frac{\hat{c}_2}{R^*} + Rq\,\hat{c}_1^{\sigma} = W$$

and $\hat{a}_2 \equiv (\hat{c}_2 - y)/R^*$. Note that $(\hat{c}_1, \hat{c}_2)$ is feasible for every $\Gamma > 0$ since $z = 0$ there.

Step 1. Set $c_2 = c_1 x^{1/\sigma}$ and $z(x) = \frac{R}{R^*}x - 1$. Define

$$H_1(c_1, x; \Gamma) \equiv c_1 + \frac{c_1 x^{1/\sigma}}{R^*} + \frac{z(x)^2}{\Gamma} + Rq\,c_1^{\sigma} - W = 0$$

and

$$H_2(c_1, x; \Gamma) \equiv Rx - \frac{F_{c_1}}{F_{c_2}} = 0$$

where the second equation is the tangency condition at an interior optimum. Then $(c_1^*(\Gamma), x^*(\Gamma))$ is the unique solution to $H_1 = H_2 = 0$.

Step 2 ($z^*(\Gamma) > 0$ for all $\Gamma > 0$). Suppose $z^*(\Gamma) = 0$ at some $\Gamma > 0$. Then $x^*(\Gamma) = R^*/R$ and the tangency equation reduces to

$$Rx^*(\Gamma) = \frac{1 + Rq\,\sigma(c_1^*(\Gamma))^{\sigma-1}}{1/R^*} = R^*\big(1 + Rq\,\sigma(c_1^*(\Gamma))^{\sigma-1}\big) > R^*$$

which contradicts $Rx^*(\Gamma) = R^*$. Hence $z^*(\Gamma) > 0$ for all $\Gamma > 0$, and therefore $x^*(\Gamma) > R^*/R$.

Step 3 ($x^*(\Gamma)$ is strictly increasing). Let $T(c_1, x; \Gamma) \equiv F_{c_1}/F_{c_2}$ evaluated at $(c_1, c_2) = (c_1, c_1 x^{1/\sigma})$. Direct differentiation of F gives

$$F_{c_1} = 1 + Rq\,\sigma c_1^{\sigma-1} - \frac{2\sigma}{\Gamma}\frac{R}{R^*}z(x)\frac{x}{c_1}, \qquad F_{c_2} = \frac{1}{R^*} + \frac{2\sigma}{\Gamma}\frac{R}{R^*}z(x)\frac{x}{c_2}$$

For any (c_1, x) with $z(x) > 0$ one has:

- $T_\Gamma > 0$ because increasing Γ raises F_{c_1} and lowers F_{c_2};
- $T_{c_1} > 0$ because increasing c_1 raises F_{c_1} and lowers the $1/c_i$ terms;
- $T_x < 0$ on $x \geqslant R^*/R$ when $\sigma \geqslant 1$ since $x \mapsto z(x)x$ and $x \mapsto z(x)x^{1-1/\sigma} = \frac{R}{R^*}x^{2-1/\sigma} - x^{1-1/\sigma}$ are strictly increasing on that region, so F_{c_1} decreases and F_{c_2} increases in x.

Hence for $H_2(c_1, x; \Gamma) = Rx - T(c_1, x; \Gamma)$ we have

$$H_{2,\Gamma} < 0, \qquad H_{2,c_1} < 0, \qquad H_{2,x} > 0$$

From H_1 we have

$$H_{1,c_1} > 0, \qquad H_{1,x} > 0, \qquad H_{1,\Gamma} < 0 \quad \text{(since } z(x)^2/\Gamma \text{ enters with a negative derivative in } \Gamma\text{)}$$

Therefore the Jacobian determinant of (H_1, H_2) satisfies

$$\det J = H_{1,c_1} H_{2,x} - H_{1,x} H_{2,c_1} > 0$$

By the implicit function theorem the solution $(c_1^*(\Gamma), x^*(\Gamma))$ is differentiable in Γ, and Cramer's rule yields

$$\frac{dx^*}{d\Gamma} = \frac{H_{2,c_1} H_{1,\Gamma} - H_{1,c_1} H_{2,\Gamma}}{\det J} > 0$$

because $H_{2,c_1} H_{1,\Gamma} > 0$ and $-H_{1,c_1} H_{2,\Gamma} > 0$. Hence $x^*(\Gamma)$ is strictly increasing in Γ.

Step 4 (limits of $x^*(\Gamma)$). Since the constraint binds and all terms of F are nonnegative, $z^*(\Gamma)^2/\Gamma \leqslant W$, so $z^*(\Gamma) \to 0$ as $\Gamma \downarrow 0$ and thus

$$\lim_{\Gamma \downarrow 0} x^*(\Gamma) = \frac{R^*}{R}$$

Next, since $(\hat{c}_1, \hat{c}_2)$ is feasible for all Γ, we have $U(c_1^*(\Gamma), c_2^*(\Gamma)) \geqslant U(\hat{c}_1, \hat{c}_2)$ for all Γ. Also $c_2^*(\Gamma) \leqslant R^* W$ from the binding constraint, hence $u(c_2^*(\Gamma)) \leqslant u(R^* W)$. Choose $\delta > 0$ such that $u(\delta) + \beta u(R^* W) < U(\hat{c}_1, \hat{c}_2)$. Then $c_1^*(\Gamma) \geqslant \delta$ for all Γ, otherwise the objective would fall below $U(\hat{c}_1, \hat{c}_2)$, contradicting optimality. Consequently $x^*(\Gamma) = (c_2^*(\Gamma)/c_1^*(\Gamma))^\sigma$ is bounded above by $(R^* W/\delta)^\sigma$. Since $x^*(\Gamma)$ is strictly increasing and bounded, it converges to a finite limit x^∞ as $\Gamma \to \infty$. Finally $x^\infty > R^*/R$ because $x^*(\Gamma) > R^*/R$ for all $\Gamma > 0$ (Step 2).

Step 5 ($c_2^*(\Gamma)$ is strictly increasing and $c_2^\infty > \hat{c}_2$). Fix $\Gamma' > \Gamma$. Because $z^*(\Gamma) > 0$, holding (c_1, c_2) fixed at $(c_1^*(\Gamma), c_2^*(\Gamma))$ strictly decreases $F(c_1, c_2; \Gamma)$ when Γ increases, so

$$F(c_1^*(\Gamma), c_2^*(\Gamma); \Gamma') < F(c_1^*(\Gamma), c_2^*(\Gamma); \Gamma) = W$$

Hence $(c_1^*(\Gamma), c_2^*(\Gamma))$ is feasible at Γ'.

Suppose, by contradiction, that $c_2^*(\Gamma') \leqslant c_2^*(\Gamma)$. If $c_1^*(\Gamma') \geqslant c_1^*(\Gamma)$ then $x^*(\Gamma') \leqslant x^*(\Gamma)$, contradicting Step 3. If $c_1^*(\Gamma') < c_1^*(\Gamma)$ then strict monotonicity of utility implies $U(c_1^*(\Gamma'), c_2^*(\Gamma')) < U(c_1^*(\Gamma), c_2^*(\Gamma))$, contradicting optimality at Γ' since $(c_1^*(\Gamma), c_2^*(\Gamma))$ is feasible there. Thus $c_2^*(\Gamma') > c_2^*(\Gamma)$, so $c_2^*(\Gamma)$ is strictly increasing.

Since $c_2^*(\Gamma) \leqslant R^* W$, it converges to a finite limit c_2^∞ as $\Gamma \to \infty$. Moreover, $c_2^*(\Gamma) > \hat{c}_2$ for all $\Gamma > 0$: if

$c_2^*(\Gamma) \leqslant \hat{c}_2$, then

$$c_1^*(\Gamma) = \frac{c_2^*(\Gamma)}{x^*(\Gamma)^{1/\sigma}} < \frac{\hat{c}_2}{(R^*/R)^{1/\sigma}} = \hat{c}_1$$

because $x^*(\Gamma) > R^*/R$, and therefore $U(c_1^*(\Gamma), c_2^*(\Gamma)) < U(\hat{c}_1, \hat{c}_2)$, contradicting optimality. Hence $c_2^\infty > \hat{c}_2$.

Step 6 (limits of $c_1^*(\Gamma)$ and $a_2(\Gamma)$). By definition $c_1^*(\Gamma) = c_2^*(\Gamma)/x^*(\Gamma)^{1/\sigma}$. Combining the limits from Steps 4 and 5 gives

$$\lim_{\Gamma \downarrow 0} c_1^*(\Gamma) = \hat{c}_1, \qquad \lim_{\Gamma \to \infty} c_1^*(\Gamma) = c_1^\infty \equiv \frac{c_2^\infty}{(x^\infty)^{1/\sigma}}$$

To compare c_1^∞ and $\hat{c}_1$, note that at $\Gamma = \infty$ the binding constraint becomes

$$c_1 + \frac{c_1 (x^\infty)^{1/\sigma}}{R^*} + Rq\, c_1^\sigma = W$$

Evaluating this expression at $c_1 = \hat{c}_1$ yields a value strictly greater than W because $x^\infty > R^*/R$. Since the left-hand side is strictly increasing in c_1, it follows that $c_1^\infty < \hat{c}_1$.

Finally, $q_2(\Gamma) = z^*(\Gamma)/\Gamma \to 0$ as $\Gamma \to \infty$ because $z^*(\Gamma) \to z^\infty = \frac{R}{R^*} x^\infty - 1 < \infty$. Hence

$$a_2(\Gamma) = \frac{c_2^*(\Gamma) - y}{R^*} + \frac{R}{R^*} x^*(\Gamma)\, q_2(\Gamma) \longrightarrow \frac{c_2^\infty - y}{R^*} \equiv a_2^\infty$$

Moreover $a_2^\infty > \hat{a}_2$ since $a_2^\infty - \hat{a}_2 = (c_2^\infty - \hat{c}_2)/R^* > 0$. □

B.4 Proof (Sketch) of Proposition 3

Proof. (Sketch) Consider the problem (24). The first order conditions for a_1, c_1, with Lagrange multipliers $(\lambda_0, \lambda_1, \lambda_2)$ on the BoP constraints and (ν_1, ν_2) on the non-negativity constraints are given below. FOCs for c_0, c_2, a_2 are omitted here for conciseness.

a_1:

$$\lambda_0 = \beta R^* \lambda_1 + \nu_1 \tag{33}$$

c_1:

$$
\begin{aligned}
\omega c_1^{-\sigma} - \lambda_1 - \lambda_1 \frac{\sigma}{c_1} R \frac{c_1^\sigma}{c_0^\sigma} \frac{1}{\Gamma} \left[\frac{R}{R^*} \left(\frac{c_1^\sigma}{c_0^\sigma} \right) - 1 \right] \\
-\lambda_1 \frac{\sigma}{c_1} \frac{1}{\Gamma} \frac{R}{R^*} \left(\frac{c_2}{c_1} \right)^\sigma \\
+\lambda_2 \frac{\sigma}{c_1} \left(\frac{c_2}{c_1} \right)^\sigma \frac{1}{\Gamma} \left[\frac{R}{R^*} \left(\frac{c_2}{c_1} \right)^\sigma - 1 \right] \\
+\lambda_2 \frac{\sigma}{c_1} \left(\frac{c_2}{c_1} \right)^\sigma \left(\frac{1}{\Gamma} \frac{R}{R^*} \left(\frac{c_2}{c_1} \right)^\sigma \right) \\
\underbrace{-R \frac{1}{\Gamma} \frac{R}{R^*} \frac{\sigma}{c_1} \left(\frac{c_1}{c_0} \right)^\sigma \left(\lambda_1 \left(\frac{c_1}{c_0} \right)^\sigma - \lambda_0 \right)}_{A}
\end{aligned}
\tag{34}
$$

The expression labeled A in (34) represents the net marginal effect of c_1 on the quantity of borrowing in period 1, q_1. The marginal benefit of additional borrowing at $t = 0$ is given by $R \frac{1}{\Gamma} \frac{R}{R^*} \frac{\sigma}{c_1} \left(\frac{c_1}{c_0} \right)^\sigma \lambda_0$. However, additional borrowing also increases the debt burden at $t = 1$. The marginal cost associated with this additional debt burden is given by $-\lambda_1 R \left(\frac{c_1}{c_0} \right)^\sigma \frac{1}{\Gamma} \frac{R}{R^*} \frac{\sigma}{c_1} \left(\frac{c_1}{c_0} \right)^\sigma$. The net marginal cost is thus given by the expression A. To show that the net effect is indeed costly at the margin, it needs to be established that A<0 at the optimum.

Since $a_1^c > 0$ by assumption, $\nu_1 = 0$. Substitute (33) into the above expression to get:

$$
\begin{aligned}
A &= -R \frac{1}{\Gamma} \frac{R}{R^*} \frac{\sigma}{c_1^c} \left(\frac{c_1^c}{c_0^c} \right)^\sigma \left(\lambda_1^c \left(\frac{c_1^c}{c_0^c} \right)^\sigma - \frac{R^*}{R} \lambda_1^c \right) \\
&= -\lambda_1^c R \left(\frac{c_1^c}{c_0^c} \right)^\sigma \frac{\sigma}{c_1^c} \frac{1}{\Gamma} \left[\frac{R}{R^*} \left(\frac{c_1^c}{c_0^c} \right)^\sigma - 1 \right] \\
&= -\lambda_1^c R \left(\frac{c_1^c}{c_0^c} \right)^\sigma \frac{\sigma}{c_1^c} q_1^c < 0
\end{aligned}
$$

Since $q_1^c > 0$ by assumption and the Lagrange multiplier on the BoP constraint, $\lambda_1^c > 0$ then $A < 0$. The idea here is that as long as $a_1 > 0$ it is always cheaper to raise resources by reducing a_1 rather than by increasing q_1 due to the resource losses involved in using the intermediation technology. Therefore, the net effect of a marginal increase in c_1 on q_1 is costly at the margin.

Now consider the first-order condition for c_1 for the two-period sub-problem (20). The FOC is identical to (34), except for the fact that the expression A now disappears because q_1 is a state variable in (20): changing c_1 does not affect q_1. Intuitively, for a time-0 planner, there is an additional marginal cost associated with increasing c_1 whereas for the time-1 planner, this cost has disappeared. A lower marginal cost implies that the time-1 planner would like to deviate to a higher level of consumption, $c_1^D > c_1^c$. Given the discussion on the two-period model, in section 4.1.1, this deviation implies a movement along the IRC from left to right, i.e., $a_2^D < a_2^c$, $q_2^D < q_2^c$ and $c_2^D < c_2^c$. □

B.5 Proof of Lemma 1

Proof. From Lemma 3, it is known that given $a_1^c, a_2^c, q_1^c > 0$ the solution to the two period problem (20) is given by:

$$c_1^D \equiv \mathcal{C}_1(c_0^c, a_1^c, q_1^c) > c_1^c$$

But since:

$$q_1^c = \frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{c_1^c}{c_0^c}\right)^{\sigma} - 1\right]$$

Then:

$$q_1^c < \frac{1}{\Gamma}\left[\frac{R}{R^*}\left(\frac{\mathcal{C}_1(c_0^c, a_1^c, q_1^c)}{c_0^c}\right)^{\sigma} - 1\right]$$

Therefore (c_0^c, a_1^c, q_1^c) violates the asset supply equation constraint for (25) and is thus infeasible. □

C Numerical Examples

In this section, we present numerical simulations illustrating the paper's main results.

C.1 Comparative Statics with Respect to Γ

Figure 7 illustrates how c_1, c_2, and a_2 change with respect to Γ in the two-period problem (20), for a given initial state (c_0, a_0, q_0). It shows that as Γ increases, c_1 decreases with Γ, reflecting the fact that the central bank allows for a larger devaluation of inherited debt in response to tighter intermediation conditions. This allows the economy to consume more in the second period, c_2, and therefore c_2 is increasing in Γ. Finally, the figure also shows that a_2 is non-decreasing in Γ and strictly increasing when the non-negativity constraint is not binding, reflecting that the central bank retains more reserves in response to tighter intermediation conditions, to allow for a larger devaluation of inherited debt.

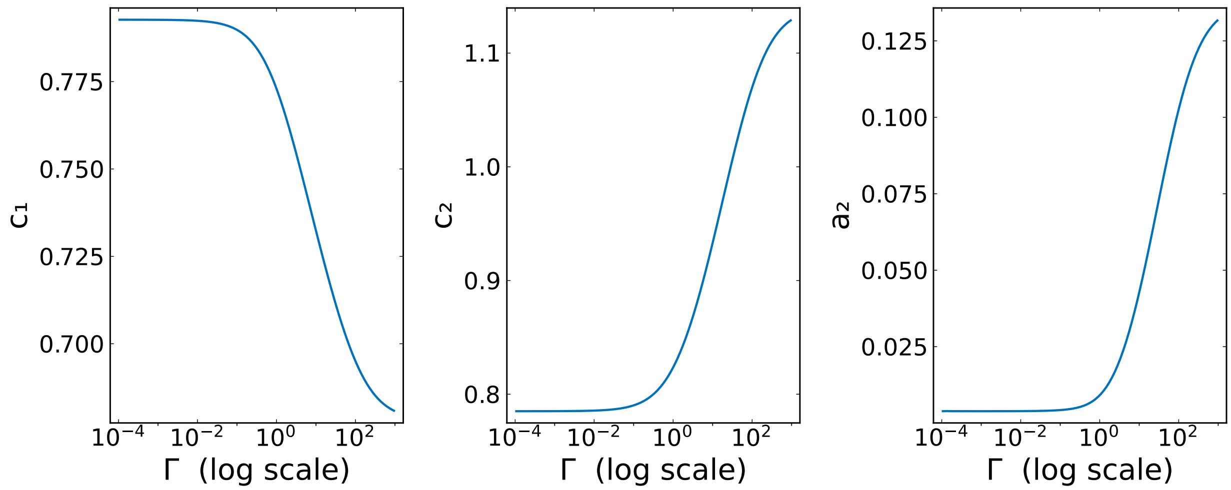

Figure 7: Comparative statics with respect to Γ

C.2 Infinite Horizon Ramsey Optimal Policy

We present a numerical example illustrating the Ramsey optimal policy for the infinite horizon policy problem discussed in Section 4.1.4. The example numerically illustrates the optimal policy path and shows that the trade-off between intermediation cost minimization and avoiding losses due to initial debt revaluation, that we analyzed in a two-period setting in Section 4.1.1, extends to the infinite horizon case. Figure 8 shows the long-run path of the equilibrium variables (c_t, a_t, q_t) chosen by the Ramsey planner given an initial state with $a_0 > 0, q_0 > 0$. As discussed in section 4.1.4, since $\beta R^* < 1$, the relative impatience implies that in the long-run, the planner eventually runs down the initial stock of reserves to zero and the economy converges to its long-run steady state with the transition dynamics for (c_t, q_t) identical to the laissez-faire dynamics described in Appendix A.

However, the short-run path of optimal policy depends on the initial debt levels and the trade-off between intermediation losses and initial debt revaluation remains operative. Figure 9 illustrates comparative statics with respect to the initial debt level q_0. It plots the interest parity gaps given by $\left(R\left(\frac{c_{t+1}}{c_t}\right)^{\sigma} - R^*\right)$ for three different initial debt levels. It shows that, all else equal, with higher initial debt levels, the planner accepts higher interest parity gaps and therefore larger intermediation costs in the short run as revaluation costs are increasing in the initial debt levels. In other words, the planner has a stronger incentive to appreciate the real exchange rate at $t = 0$ when debt levels are higher.

Finally, Figure 10 directly compares the short-run paths of optimal policy against a naive policy that minimizes the carry cost by netting out the country's balance sheet at t = 0. In this example, the planner immediately runs down reserves to zero at t = 0 to minimize the country's external debt position and therefore minimize intermediation costs. This policy, which immediately liquidates all reserves, induces a stronger real exchange rate appreciation relative to optimal policy: the household enjoys higher consumption in the initial few periods. By contrast, the optimal policy retains reserves and slowly runs them down over time. Due to this reserve retention, optimal policy features larger household borrowing and larger interest parity gaps in the initial periods. However, the naive carry-cost minimization policy induces a strong real exchange rate appreciation at t = 0 which leads to a revaluation of initial debt that shows up in higher ex post profits, ν_0, for the intermediaries. The figure also plots the present discounted value of intermediary profits cumulatively. These represent the overall intermediation premia paid by the economy over time including the time-0 profits associated with previously contracted debt. The figure shows that the carry cost minimization policy creates a larger resource loss for the economy, much of that loss happens at t = 0 due to the large revaluation of initial debt. By contrast, the optimal policy minimizes the overall resource loss by accepting higher current and future profits against lower time-0 profits.

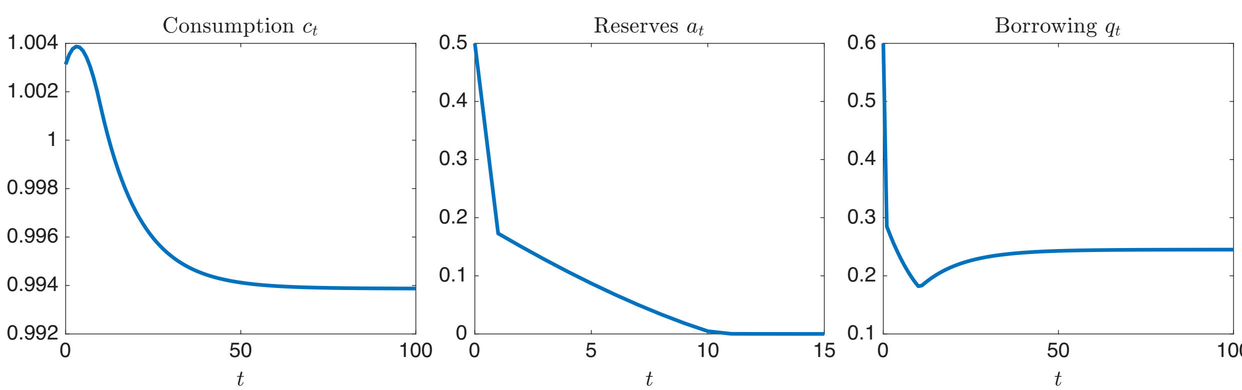

Figure 8: Optimal policy: long-run dynamics

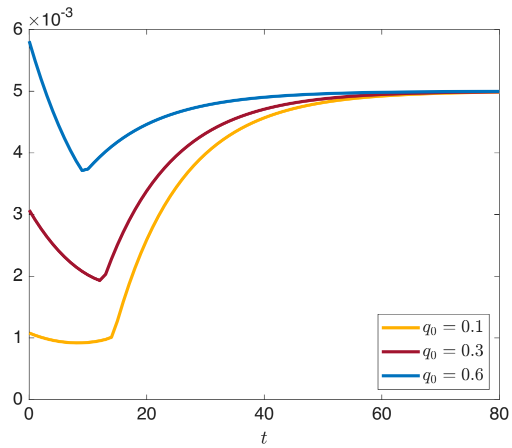

Figure 9: Interest parity gaps: comparative statics w.r.t. q_0

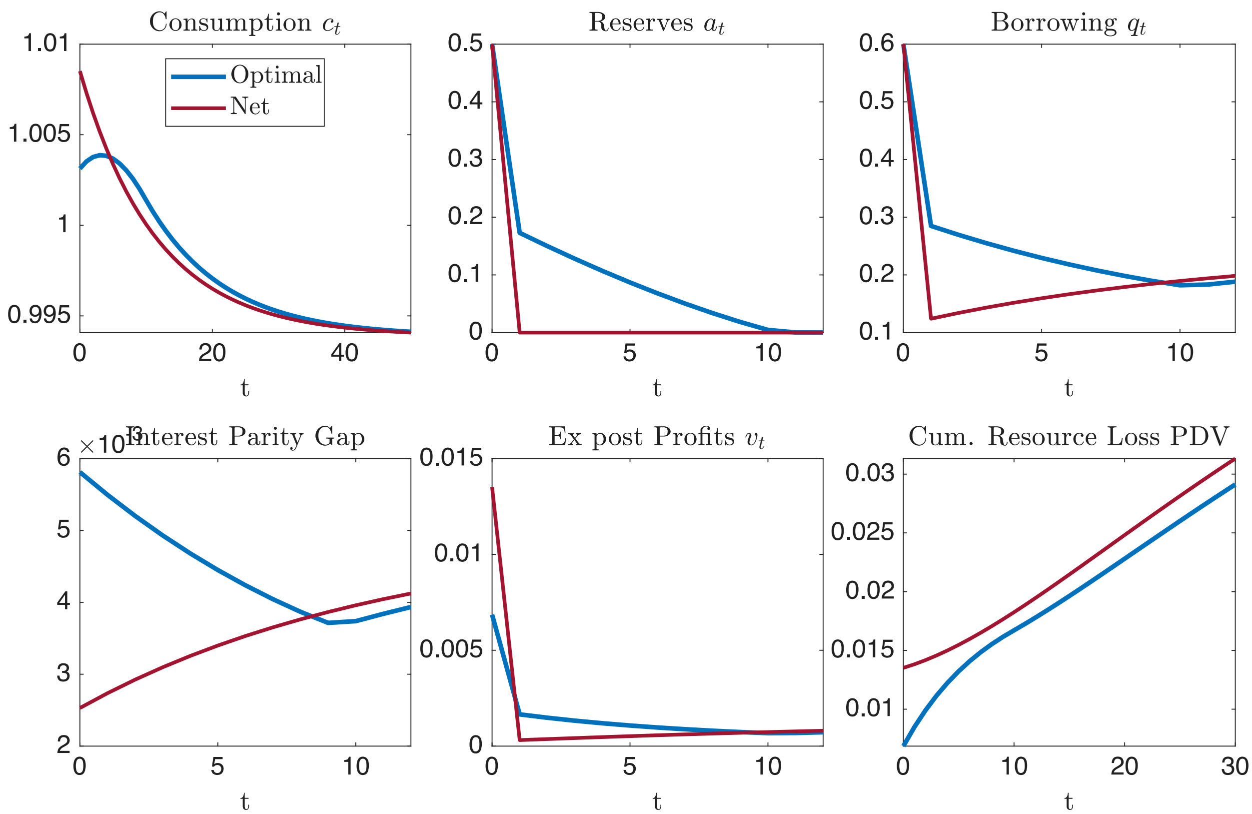

Figure 10: Optimal policy vs carry cost minimization

C.3 Commitment vs Discretion

Figure 11 illustrates the time-inconsistency of Ramsey optimal policy. First it plots the optimal policy given a state at $t = 0$ with $a_0 > 0, q_0 > 0$. Then it plots a deviation policy at $t = 1$ where the planner is allowed to re-optimize at $t = 1$ given the time-0 planner's choices, taking $a_1, \frac{q_1}{c_0^{\sigma}}$ as state variables. The figure shows that at $t = 1$ the deviating planner prefers a stronger real exchange rate, i.e., a higher level of consumption than the commitment plan originally prescribed. This deviation is consistent with the intuition from the three-period model that the time-1 planner faces a lower marginal cost of increasing c_1 than the time-0 planner, and therefore prefers a higher level of consumption at $t = 1$.

Figure 12 compares the time-consistent equilibrium with the Ramsey optimal 'commitment' policy in the three-period model. The figure shows, given an identical initial state, the economy features higher reserves ($a_1^l > a_1^c$) and a larger interest parity gap ($q_1^l > q_1^c$) in the time-consistent equilibrium relative to the commitment equilibrium at $t = 0$. Since intermediaries anticipate that the planner will deviate to a higher level of consumption at $t = 1$, they lend more at $t = 0$ and the central bank accumulates more reserves to prevent an excessive appreciation and the resulting revaluation of initial debt. The figure also illustrates the relative impatience that lack of commitment generates relative to the commitment equilibrium. A commitment plan chooses a higher level of consumption at $t = 2$ but the time-consistent policy chooses higher consumption at $t = 0, 1$ and lower consumption at $t = 2$.

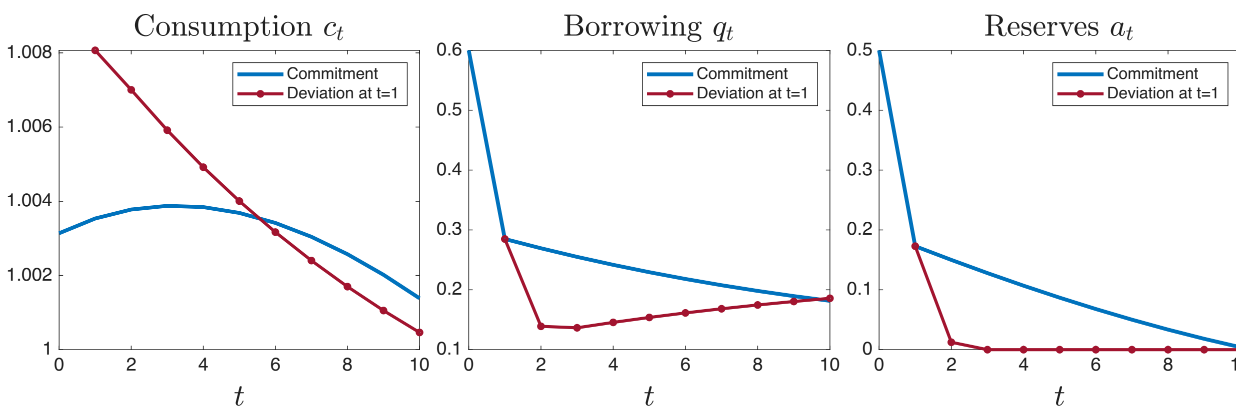

Figure 11: Proposition 4 illustrated

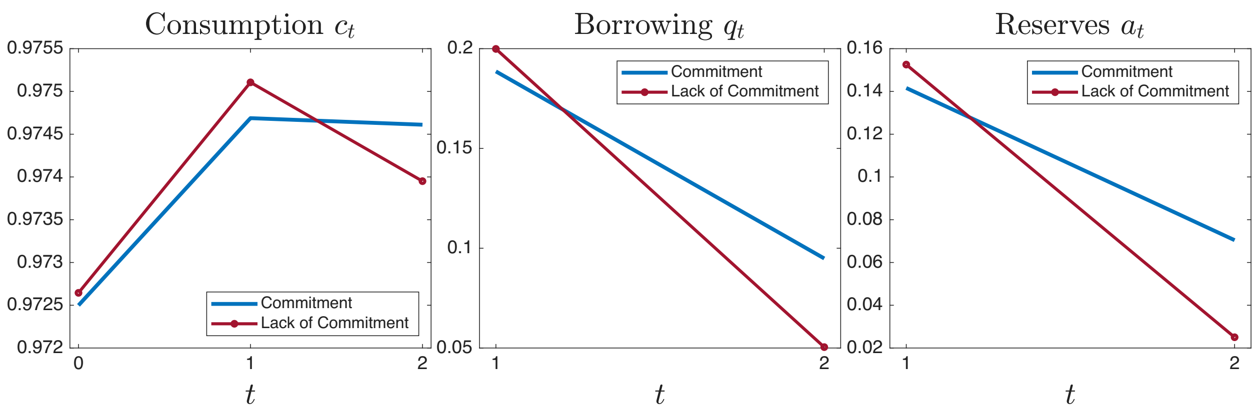

Figure 12: Proposition 5 illustrated

C.4 Time-Consistent Equilibrium: Policy Functions and Revaluation Costs

Figures 13 and 14 plot slices of the policy functions for debt and the exchange rate respectively as a function of the initial level of reserves and local currency debt, for the low and high states of Γ.

Figure 15 plots the marginal cost of reserve deployment in terms of debt revaluation. We show that this debt revaluation cost is strictly increasing in the tightness of financial intermediation. To formalize this, let $\hat{e}(a, \tilde{q}, \Gamma; a')$ be the locus of exchange rate values given a state, future policies, and a reserve choice a' that satisfies the constraints of (26). At the optimum, the real payments to foreigners are given by:

$$P = \frac{R\tilde{q}}{\hat{e}(a, \tilde{q}, \Gamma; a')}\bigg|_{a'=\mathcal{A}(a,\tilde{q},\Gamma)}$$

The marginal cost of reserve deployment can be expressed as the decrease in real payments to foreigners when the central bank marginally increases reserves at the optimum and induces a depreciation:

$$\Delta = -\frac{\partial P}{\partial a'}\bigg|_{a'=\mathcal{A}(a,\tilde{q},\Gamma)} = \frac{R\tilde{q}}{\hat{e}(a, \tilde{q}, \Gamma; a')^2}\frac{\partial \hat{e}(a, \tilde{q}, \Gamma; a')}{\partial a'}\bigg|_{a'=\mathcal{A}(a,\tilde{q},\Gamma)}$$

Not only is this cost increasing in the level of initial debt, $\tilde{q}$, but it is also increasing in the severity of financial disruption, Γ. This is illustrated in Figure 15 where this marginal cost is plotted as a function of the initial level of local currency debt, $\tilde{q}$, for the mid and high states of Γ, i.e., the states where tradeable resources are relatively scarce and the central bank deploys reserves.

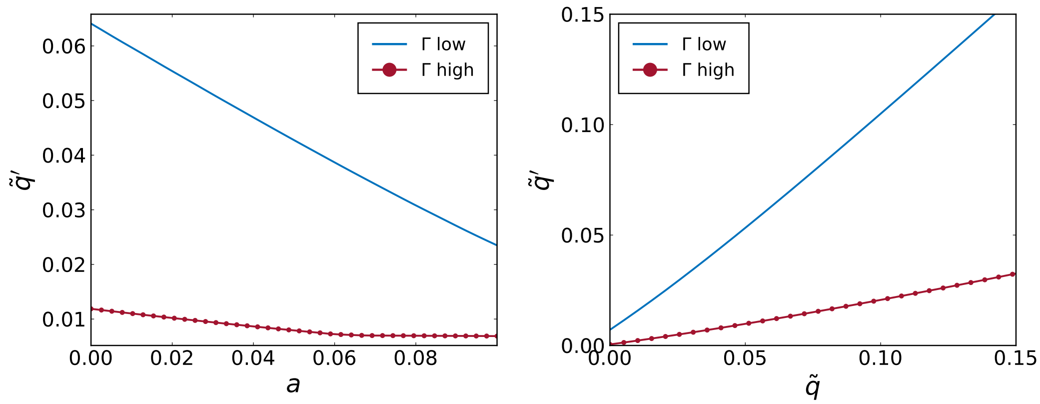

Figure 13: Policy Function- Debt

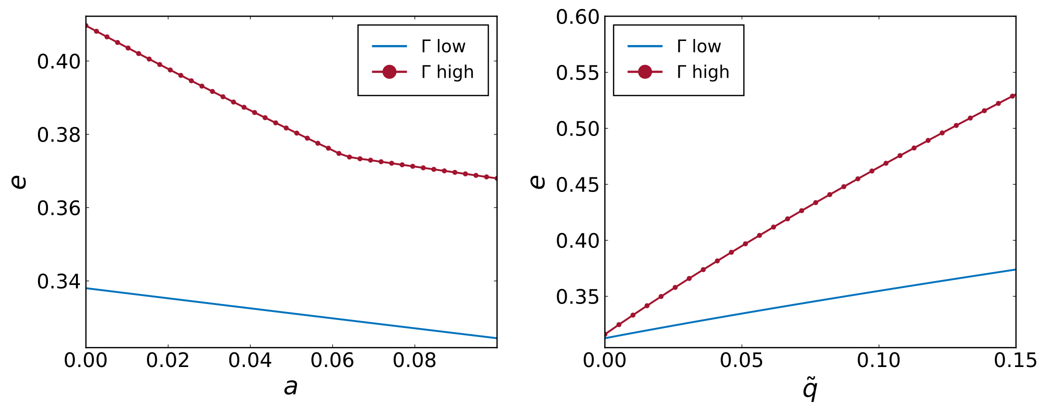

Figure 14: Policy Function- Exchange Rate

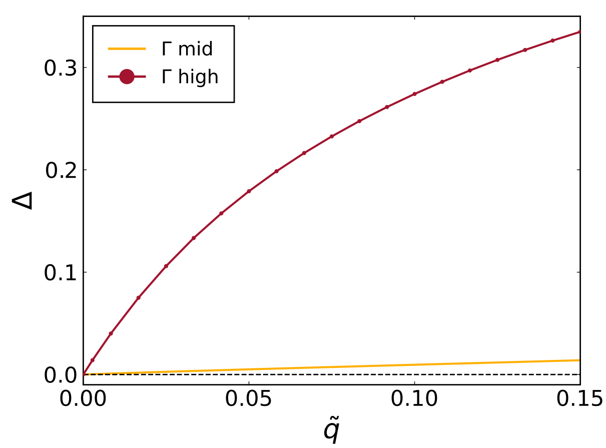

Figure 15: Marginal cost of reserve deployment- debt revaluation